

1 Three-component decomposition of treatment response  
2 in aggregated N-of-1 trials: pharmacological,  
3 expectation-driven, and natural-history components

4 pmsimstats team

5 2026-06-15

6 **Contents**

|    |                                                                            |           |
|----|----------------------------------------------------------------------------|-----------|
| 7  | <b>Abstract</b>                                                            | <b>2</b>  |
| 8  | <b>1 Introduction</b>                                                      | <b>3</b>  |
| 9  | 1.1 A claim about how N-of-1 trials should be analysed . . . . .           | 3         |
| 10 | 1.2 Why this matters: three failure modes of the lumped analysis . . .     | 4         |
| 11 | 1.3 The current state of the N-of-1 literature . . . . .                   | 8         |
| 12 | 1.4 The neurobiological and epidemiological reality of the components      | 11        |
| 13 | 1.5 What this paper contributes . . . . .                                  | 13        |
| 14 | <b>2 Notation and the formal model</b>                                     | <b>13</b> |
| 15 | <b>3 The three components</b>                                              | <b>16</b> |
| 16 | 3.1 Biological response (BR) . . . . .                                     | 16        |
| 17 | 3.2 Placebo-belief response (PB) . . . . .                                 | 20        |
| 18 | 3.3 Time-variant natural-history component (TV) . . . . .                  | 29        |
| 19 | <b>4 Identifiability and trial-design contrasts</b>                        | <b>34</b> |
| 20 | <b>5 Analysis-side methodology</b>                                         | <b>37</b> |
| 21 | 5.1 Four reasons not to match the DGP . . . . .                            | 38        |
| 22 | 5.2 A small extension that usually pays off: phase indicators . . . . .    | 41        |
| 23 | 5.3 When to fit the full decomposition . . . . .                           | 42        |
| 24 | <b>6 Worked example: information loss without decomposition</b>            | <b>44</b> |
| 25 | 6.1 What the lumped analysis estimates . . . . .                           | 44        |
| 26 | 6.2 A numerical illustration . . . . .                                     | 47        |
| 27 | <b>7 Simulation study</b>                                                  | <b>53</b> |
| 28 | 7.1 Pilot validation of analysis machinery (100 replicates, 6 cells) . . . | 53        |
| 29 | 7.2 Production-simulation specification . . . . .                          | 55        |

|    |                                                                         |           |
|----|-------------------------------------------------------------------------|-----------|
| 30 | 7.3 Study A results: bias and power at 1,000 replicates per alternative |           |
| 31 | cell . . . . .                                                          | 56        |
| 32 | 7.4 Contaminated-biomarker pilot (100 replicates per cell) . . . . .    | 58        |
| 33 | <b>8 Application to predictive-biomarker validation</b>                 | <b>61</b> |
| 34 | <b>9 Discussion</b>                                                     | <b>63</b> |
| 35 | <b>10 Conclusions</b>                                                   | <b>67</b> |
| 36 | <b>11 Conflict of interest</b>                                          | <b>69</b> |
| 37 | <b>12 Funding</b>                                                       | <b>69</b> |
| 38 | <b>13 Data availability statement</b>                                   | <b>69</b> |
| 39 | <b>14 Reproducibility</b>                                               | <b>69</b> |
| 40 | <b>15 References</b>                                                    | <b>71</b> |

## 41 Abstract

42 **Background.** Treatment response in aggregated N-of-1 clinical trials reflects  
43 three causally distinct mechanisms: pharmacological action of the active com-  
44 pound (BR), the patient’s belief that the treatment will work (PB), and natural-  
45 history drift in the underlying condition (TV). Standard analysis models lump  
46 these into a single ‘treatment effect’ plus residual error. Whether this lumping  
47 biases inference is not unconditional: it depends on the estimand and on whether  
48 candidate biomarkers correlate with the non-pharmacological components, a de-  
49 pendence this paper makes precise.

50 **Methods.** We formalise a three-component decomposition of the within-  
51 participant outcome trajectory, each component a modified Gompertz function  
52 with participant-specific random effects, and characterise the trial-design con-  
53 trasts (open-label, blinded discontinuation, crossover) that supply identifiability.  
54 We derive the omitted-variable-bias identity for the one-component estimator,  
55 which expresses both the lumped treatment effect and the lumped biomarker  
56 slope as sums of component-specific terms, and we evaluate the analysis strategies  
57 in a pre-registered Monte Carlo study (Study A) calibrated to the prazosin-PTSD  
58 reference parameters, comparing a one-component design-contrast analysis with  
59 a phase-augmented analysis at 1,000 replicates per alternative cell (5,000 per null  
60 cell) across  $m_{PB} \in \{0, 1, 3, 6, 10\}$ ,  $m_{TV} \in \{-1, 0, 1, 2\}$ , and  $N \in \{35, 70, 100, 150\}$ .

61 **Results.** The identity shows that the lumped biomarker-by-treatment slope is dis-  
62 placed from the pharmacological slope by a term proportional to the biomarker’s  
63 covariance with PB and TV, not by the magnitude of those components. Study  
64 A confirms this: with a biomarker coupled to BR only, the one-component esti-  
65 mate of the biomarker-by-treatment interaction is essentially unbiased across the

whole grid (mean absolute bias 0.016 at  $N = 35$  and 0.006 at  $N = 150$ , within one Monte Carlo standard error of zero), with near-nominal coverage and type I error. The phase-augmented analysis provides no bias reduction and is markedly less powerful (mean power 0.35 versus 0.59 at  $N = 35$ ; 0.62 versus 0.90 at  $N = 70$ ), so it is dominated under this uncontaminated biomarker. The inferential value of decomposition is therefore conditional, not universal.

**Conclusions.** The trial design must supply the drug-on/off and belief contrasts; given those, component-level decomposition is useful when the estimand is the attribution of response to pharmacology versus belief versus natural history, or when a candidate biomarker may correlate with PB or TV. For a biomarker-by-treatment interaction with a biomarker orthogonal to PB and TV, a single-indicator analysis that exploits the design contrast is sufficient and a parametric component model is counterproductive. A pragmatic specification for the cases that warrant it is  $Sx \sim bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase$  with random intercept and AR(1) residual correlation. The decisive test of the biomarker-validation failure, a data-generating process that couples the biomarker to PB, remains to be run.

## 1 Introduction

### 1.1 A claim about how N-of-1 trials should be analysed

[1]

**This paper argues that treatment response in N-of-1 trials comprises three causally distinct components that must be analysed jointly rather than lumped.**

- The three components are *BR* (pharmacological), *PB* (placebo-belief), and *TV* (natural history).
- Each component is causally distinct, clinically interpretable, and identifiable from a well-designed N-of-1 trial.
- The standard single-indicator mixed-effects model conflates all three and renders biomarker validation unanswerable.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

This paper advances an opinionated methodological claim: treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components and should routinely be analysed as such, rather than estimated as a single lumped quantity. The three components are pharmacological action (*BR*, the biological response), placebo-belief response (*PB*, the patient's belief that they are receiving treatment), and time-variant natural history (*TV*, the trajectory that would have occurred without intervention). Each is causally distinct, each is independently of clinical and regulatory interest, and each is identifiable

106 from a properly designed N-of-1 trial. We shall argue that the standard  
107 practice of fitting a single mixed-effects model with one treatment indi-  
108 cator, prevalent across the published N-of-1 literature, conflates these  
109 three components, biases the interpretation of treatment effect, and  
110 renders predictive-biomarker validation unanswerable as a scientific  
111 question.

112 [2]

113 **The contribution is the consolidation and extension of an**  
114 **established structural-decomposition tradition, not a novel**  
115 **decomposition.**

- 116 • The additive disease-progression, placebo, and drug-action de-  
117 composition is well established in pharmacometrics and longitu-  
118 dinal trial methodology.
- 119 • The present framework extends that tradition with an explicit,  
120 separately identified placebo-belief channel and brings it to the  
121 aggregated N-of-1 setting.
- 122 • The decomposition logic is not specific to N-of-1; it applies wher-  
123 ever a longitudinal treatment response is analysed.

124 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
125 *tempor incididunt ut labore et dolore magna aliqua.*

126 We make no claim of novelty for the decomposition itself. The struc-  
127 tural decomposition of a longitudinal treatment response into sepa-  
128 rable disease-progression, placebo, and drug-action components is a  
129 well-established tradition in pharmacometrics<sup>1–3</sup>, and the separation  
130 of drug from placebo response in longitudinal trajectories has been  
131 pursued both through growth-mixture models<sup>4</sup> and through experi-  
132 mental decomposition of the placebo response itself<sup>5</sup>. Each ingredient  
133 we use (mixed-effects models with trial-phase indicators, placebo-arm  
134 contrasts, regression-to-mean adjustment, growth-curve parameterisa-  
135 tion) has likewise appeared across decades of clinical-trial methodol-  
136 ogy. Our contribution is therefore one of consolidation and extension  
137 rather than invention: we make the three-component structure explicit  
138 for the aggregated N-of-1 setting, add a separately modelled placebo-  
139 belief channel identified by design contrasts rather than by functional  
140 form alone, and argue that the decomposition should become routine  
141 wherever a longitudinal treatment response is analysed. We document  
142 why the framing has not been routine in N-of-1 practice (parsimony,  
143 identifiability concerns, sample-size constraints, methodological iner-  
144 tia) and propose an analysis that is tractable at trial-relevant sample  
145 sizes through a phase-augmented linear-mixed-effects approximation  
146 to the full decomposition.

## 147 1.2 Why this matters: three failure modes of the lumped analysis

[3]

**The lumped one-component analysis has three documented failure modes that grow with unmodelled component heterogeneity.**

- The three failure modes are placebo-attribution bias, regression-to-the-mean, and biomarker-validation failure.
- Each failure mode is documented in the chronic-condition trial literature, not merely hypothetical.
- The failures are present regardless of sample size.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

We begin with the failure modes of the standard approach. A one-component analysis (‘the treatment effect is what we observe minus baseline’) has three identifiable failure modes that arise when the population displays meaningful heterogeneity in any of the unmodelled components. None of these is hypothetical: each has a documented presence in the chronic-condition trial literature. Two points of precision, established formally in section 6.1 and borne out by the simulation of section 7.3, should be carried through what follows. The first two failure modes attach to the baseline-anchored lumped effect and are removed once the design supplies a drug-on/off contrast, which the one-component analysis of later sections in fact uses. The third, biomarker-validation failure, is different in kind: it requires not merely that  $PB$  or  $TV$  be large but that the candidate biomarker correlate with them, and where that coupling is absent the lumped analysis is not biased at all.

[4]

**Placebo-attribution bias is sample-size-invariant and documented across antidepressant, anxiety, and IBS literatures.**

- Lumped treatment-effect estimates overstate pharmacological efficacy in proportion to  $PB$  magnitude and variance.
- Kirsch et al. and Locher et al. demonstrated that antidepressant apparent efficacy is largely explained by placebo response and natural fluctuation.
- Placebo responsiveness has measurable biological correlates (e.g. COMT genotype), confirming it is a structured phenomenon, not noise.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The first is the placebo-attribution failure. When the population contains both high and low placebo responders, the lumped treatment

189 effect overestimates the pharmacological component. The bias scales  
190 with the magnitude and variance of  $PB$  in the study population, and  
191 it is invariant under sample-size increases: a larger one-component  
192 trial converges on the same biased estimate. The most consequential  
193 example in modern clinical pharmacology is antidepressant efficacy.<sup>6</sup>  
194 analysed FDA-submitted data on four widely- prescribed antidepres-  
195 sants and showed that, except at the most severe baseline-severity  
196 stratum, mean drug-placebo differences fell below clinical-significance  
197 thresholds, with the apparent ‘efficacy’ largely explained by placebo  
198 response and natural fluctuation;<sup>7</sup> reproduced this pattern in late-life  
199 depression and explicitly invoked regression to the mean.<sup>8</sup> generalised  
200 the finding to anxiety. Patient populations differ systematically in  
201 placebo responsiveness, and the difference has measurable biological  
202 correlates: COMT val158met genotype predicts placebo response in  
203 irritable bowel syndrome<sup>9,10</sup>, indicating that ‘placebo’ is not a homoge-  
204 neous nuisance but a structured biological phenomenon that interacts  
205 with the patient’s genome and treatment history. A trial whose pop-  
206 ulation happens to be placebo-responsive will report a larger ‘drug  
207 effect’ than one whose population is not, even when the underlying  
208 pharmacology is identical.

209 [5]

210 **Regression-to-the-mean contributes several points of appar-**  
211 **ent improvement in early trial weeks and is absorbed without**  
212 **trace by the lumped analysis.**

- 213 • Enrolment at peak severity guarantees a natural-history regres-  
214 sion component irrespective of pharmacology.
- 215 • Hrobjartsson et al. systematic reviews across 200+ trials showed  
216 much of conventional placebo response was indistinguishable  
217 from natural-history drift.
- 218 • The lumped analysis has no formal mechanism to recover the  
219 pharmacological effect once regression is absorbed.

220 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
221 *tempor incididunt ut labore et dolore magna aliqua.*

222 The second failure mode is regression-to-the-mean. When patients en-  
223 rol at moments of clinical engagement that coincide with peak symp-  
224 tom severity, natural-history regression toward typical severity con-  
225 tributes several points of apparent improvement in the early weeks  
226 of any trial, without any pharmacological contribution at all. The  
227 systematic reviews of<sup>11, 12</sup>, and<sup>13</sup>, the last covering 202 trials across 60  
228 clinical conditions, quantified this directly: in trials with both placebo  
229 and no-treatment arms, much of what is conventionally attributed to  
230 the placebo response was indistinguishable from regression to the mean  
231 and natural-history drift. The lumped analysis absorbs this regression

232 into the treatment-effect estimate and the analyst has no formal mech-  
233 anism to recover the true pharmacological effect.

234 [6]

235 **Biomarker validation is invalidated when a biomarker corre-**  
236 **lates with  $PB$  or  $TV$ , rendering precision-medicine stratifica-**  
237 **tion wrong regardless of trial size.**

- 238 • A biomarker correlated with placebo responsiveness or natural-  
239 history trajectory is indistinguishable from a true pharmacologi-  
240 cal predictor in a lumped analysis.
- 241 • The PATH statement defers mechanistic attribution to do-  
242 main knowledge; the BR-PB-TV decomposition supplies that  
243 substrate.
- 244 • Where such coupling cannot be ruled out, rigorous pharmacological-  
245 biomarker validation requires the decomposition; where the  
246 biomarker is known orthogonal to  $PB$  and  $TV$ , the lumped  
247 analysis already recovers the pharmacological slope.

248 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
249 *tempor incididunt ut labore et dolore magna aliqua.*

250 The third failure mode is the biomarker-validation failure. The  
251 precision-medicine question that motivates many recent N-of-1 trials  
252 is whether a baseline biomarker predicts treatment response, for-  
253 malised in the PATH statement<sup>14,15</sup>. A biomarker that correlates with  
254 placebo responsiveness or with a favourable natural-history trajectory  
255 will appear in a one-component analysis as a predictor of the lumped  
256 ‘treatment effect’, indistinguishable from a true pharmacological  
257 predictor. The biomarker-stratified prescribing decision that follows  
258 from such an analysis is wrong, and the trial’s regulatory utility  
259 is degraded. The PATH framework explicitly defers the mecha-  
260 nism of treatment-effect heterogeneity to domain knowledge<sup>16,17</sup>;  
261 the BR-PB-TV decomposition supplies the mechanistic substrate  
262 that PATH leaves blank. We argue that, whenever the candidate  
263 biomarker may itself correlate with  $PB$  or  $TV$ , predictive-biomarker  
264 validation cannot be conducted rigorously in a lumped-analysis  
265 framework regardless of how large the trial is; the bias is fixed by the  
266 biomarker’s coupling to the non-pharmacological components, not  
267 by sample size. Where the biomarker can be assumed orthogonal  
268 to  $PB$  and  $TV$ , this failure mode does not arise, and the lumped  
269 analysis recovers the pharmacological slope, as the simulation of  
270 section 7.3 confirms. We stress at the outset that this failure mode  
271 is established in this paper as an algebraic identity (section 6.1), not  
272 yet as a simulated phenomenon: the executed Study A (section 7.3)  
273 deliberately varies the *magnitude* of  $PB$  and  $TV$  with the biomarker  
274 coupled to  $BR$  alone, which the identity predicts to be innocuous,

275 while the decisive cell that couples the biomarker to  $PB$  remains to  
276 be run. The failure-mode claims of this section should therefore be  
277 read as consequences of the identity, whose empirical demonstration  
278 is a stated item of outstanding work rather than a result reported  
279 here.

### 280 1.3 The current state of the N-of-1 literature

281 [7]

282 **The N-of-1 literature has matured substantially but moti-**  
283 **vates three observations that the present paper addresses.**

- 284 • The design has evolved from individual-patient consultation to a  
285 framework supporting population-level inference.
- 286 • CONSORT reporting standards and quality audits document pro-  
287 gressive improvement in analytic transparency.
- 288 • Three specific gaps in the literature motivate the decomposition  
289 framework proposed here.

290 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
291 *tempor incididunt ut labore et dolore magna aliqua.*

292 The N-of-1 trial in its modern randomised form originates with<sup>18</sup>, who  
293 defined the within-patient multiple-crossover RCT, demonstrated it on  
294 a single airflow-limitation case, and established a clinical service to de-  
295 liver it at scale. The design inherits its identifying logic from two  
296 older traditions: the crossover trial of classical biostatistics, in which  
297 within-subject period contrasts estimate treatment effects under ex-  
298 plicit carryover assumptions, and the single-case experimental designs  
299 of behavioural science, in which phase reversals (withdrawal and rein-  
300 statement of treatment) identify effects within one participant. The  
301 blinded-discontinuation contrast central to this paper is a direct de-  
302 scendant of both. The methodological step from the individual-patient  
303 trial to population-level inference was taken by<sup>19</sup> and<sup>20</sup>, who showed  
304 that a series of single-patient trials can be combined through hierarchi-  
305 cal meta-analysis to estimate both population treatment effects and  
306 individual responses, the statistical foundation on which the aggre-  
307 gated design rests. Over the last fifteen years the design has accord-  
308 ingly been reframed from an individual-patient decision tool into an  
309 aggregated, meta-analytic framework supporting population-level in-  
310 ference: the programmatic statement of<sup>21</sup>, the methodological reviews  
311 of<sup>22</sup>, the historical-practice audit of<sup>23</sup>, and the more recent syntheses  
312 of<sup>24</sup>,<sup>25</sup>,<sup>26</sup>, and<sup>27</sup> trace that trajectory. Reporting standards are codi-  
313 fied in the CONSORT extension<sup>28</sup> (the headline statement) and<sup>29</sup> (the  
314 explanation-and-elaboration companion); the reporting-quality audits  
315 of<sup>30</sup> and its successors document slow but progressive improvement in  
316 analytic transparency. Three observations about this literature moti-

317 vate the present paper.

318 [8]

319 **No published N-of-1 trial employs a structural decomposition**  
320 **of treatment response despite three decades of methodologi-**  
321 **cal literature.**

- 322 • Audit data show the field has relied on paired t-tests, binary-  
323 indicator LME, and hierarchical Bayesian regressions.
- 324 • Recent methodological advances (Liao 2023, Blackston 2019, Car-  
325 rozzo 2025) retain the lumped-response framing.
- 326 • The decomposition gap persists even as carryover and identifica-  
327 bility problems are increasingly recognised.

328 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
329 *tempor incididunt ut labore et dolore magna aliqua.*

330 The first observation is that the dominant analytic models are sim-  
331 ple. The reporting-quality audit of<sup>30</sup> documents that the published  
332 N-of-1 trials between 1985 and 2013 relied predominantly on paired  
333 t-tests, simple linear mixed-effects models with a binary treatment in-  
334 dicator, or hierarchical Bayesian regressions. On our own reading of  
335 that corpus and the methodological literature that followed it, no pub-  
336 lished N-of-1 trial has employed a structural decomposition of treat-  
337 ment response into pharmacological, expectancy, and natural-history  
338 components. The most recent methodological advances retain this  
339 character:<sup>31</sup> develops Bayesian distributed-lag models with autocor-  
340 related errors as a current carryover-aware analysis, but the model  
341 still treats the response as a single quantity to be estimated.<sup>32</sup> and<sup>33</sup>  
342 compare aggregated N-of-1 trials with parallel and crossover RCT al-  
343 ternatives by simulation and document carryover-induced type-I-error  
344 inflation as a key failure mode, but again at the level of the lumped  
345 treatment effect. None of these papers decompose the response.

346 [9]

347 **Placebo-component separation is widely recognised but al-**  
348 **most entirely unaddressed at the protocol level in N-of-1**  
349 **methodology.**

- 350 • The Hendrickson hybrid blinded-discontinuation phase is among  
351 the few protocol mechanisms that supply *PB*-versus-*BR* identi-  
352 fiability.
- 353 • Placebo run-in, open-label-placebo, and mediator-analysis  
354 approaches have not been imported into N-of-1 designs.
- 355 • The gap between awareness of the problem and protocol-level  
356 solutions motivates the present framework.

357 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*

358 *tempor incididunt ut labore et dolore magna aliqua.*

359 The second observation is that the placebo-component separation  
360 problem is widely known but largely unaddressed in N-of-1 specifically.  
361 The blinded discontinuation phase used in the Hendrickson hybrid  
362 design<sup>34</sup> is one of the few protocol-level mechanisms in routine N-of-1  
363 use that supplies identifiability of *PB* versus *BR*. Other relevant  
364 approaches, including placebo run-in designs, open-label-placebo  
365 trials, and mediator-analysis applications, have been developed in  
366 parallel but rarely imported into N-of-1 methodology.

367 [10]

368 **Kaptchuk’s within-placebo decomposition is the closest**  
369 **structural analogue; no published paper applies the corre-**  
370 **sponding full-response decomposition to N-of-1 trials.**

- 371 • Kaptchuk et al. (2008) empirically decomposed the placebo re-  
372 sponse into observation, ritual, and relationship components.
- 373 • Open-label placebo trials confirm that clinically meaningful PB-  
374 only response is possible even when patients know they receive  
375 placebo.
- 376 • The BR-PB-TV framework is structurally analogous but operates  
377 at the level of the full treatment response.

378 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
379 *tempor incididunt ut labore et dolore magna aliqua.*

380 The third observation is that the closest precedent in the literature is  
381 Kaptchuk’s three-component decomposition of the placebo response  
382 itself. In a randomised trial of acupuncture-style placebo for irritable  
383 bowel syndrome,<sup>5</sup> decomposed the placebo response into observation,  
384 ritual, and patient-practitioner relationship components, demonstrat-  
385 ing empirically that the placebo response is itself a sum of identifiable  
386 parts rather than a single ‘expectancy’ lump. The follow-up open-label  
387 placebo trials of<sup>35, 36</sup>, and<sup>37</sup> in adolescent functional pain establish that  
388 even disclosed-placebo intervention can produce clinically meaningful  
389 PB-only response. The framework we propose is structurally anal-  
390 ogous to Kaptchuk’s decomposition but applied at the level of the  
391 full treatment response (drug, placebo, natural history) rather than  
392 within the placebo response alone. To the best of our knowledge, no  
393 published paper proposes the corresponding decomposition for the ag-  
394 gregated N-of-1 setting, even though the methodological ingredients,  
395 including mixed-effects modelling, blinded-phase contrasts, and trajec-  
396 tory parameterisation, are all in routine use.

397 [11]

398 **The decomposition gap is real, the tools to fill it are ma-**  
399 **ture, and the failure modes are large enough to make it a**

**methodological priority.**

- The gap between available methodology and current practice is documented and not attributable to missing tools.
- Failure modes in chronic-condition pharmacology are consequential, not academic.
- The remainder of the paper develops, applies, and evaluates the decomposition framework.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The case we develop in the rest of this paper is therefore that the gap is real, that the methodological tools to fill it are mature, and that the failure modes of the lumped analysis are large enough in chronic-condition pharmacology to make the decomposition a methodological priority rather than an academic refinement.

## **1.4 The neurobiological and epidemiological reality of the components**

[12]

**Each component of the decomposition reflects a real causal mechanism documented in independent literatures.**

- *BR* corresponds to pharmacological action; *PB* to neurobiologically real belief-driven mechanisms; *TV* to natural-history drift.
- The components are not statistical constructs but distinct causal pathways with separate literatures.
- The section reviews the evidence for each component before establishing the formal model.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The decomposition is not a statistical convenience. Each component reflects a real causal mechanism documented in independent literatures, and we shall briefly recall each in turn before turning to the formal model.

[13]

***BR* is the pharmacological target of regulation; *PB* is neurobiologically real and phase-modulated, which is precisely the design contrast that makes the decomposition identifiable.**

- *PB* is mediated by endogenous opioid/dopamine release, HPA-axis regulation, and top-down attentional shifts.
- Wager et al. neurologic pain signature provides a clean fMRI dissociation between pharmacological and expectancy contributions.

- Phase-dependent belief variation (open-label vs. blinded) is the design contrast that supplies *PB* identifiability.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The biological response component represents the pharmacological action of the active compound: the chemical binding, downstream signalling, and physiological response that a regulatory authority cares about when it grants approval. The placebo-belief response reflects neurobiologically real mechanisms documented across pain, depression, Parkinson's, and PTSD research<sup>38–41</sup>, mediated by belief-driven release of endogenous opioids and dopamine, anticipatory regulation of the hypothalamic-pituitary-adrenal axis, and top-down attentional shifts in symptom perception. Direct neuroimaging evidence is instructive here:<sup>42</sup> developed an fMRI-based 'neurologic pain signature' (NPS) that tracks nociceptive input and is reduced by the opioid agonist remifentanyl, and the individual-participant meta-analysis of<sup>43</sup> found that placebo treatments produce moderate reductions in reported pain but only negligible reductions in the NPS, providing a neural dissociation between the pharmacological and expectancy contributions to a clinical pain response. The placebo response is heterogeneous across patients with biological correlates<sup>9,10</sup> and is modulated by trial phase: when belief varies, as it does between open-label and blinded-discontinuation conditions, the placebo response varies, and the modulation is exactly the design contrast that makes the decomposition identifiable.

[14]

***TV* is the untreated counterfactual trajectory; its heterogeneity is the central object of precision-medicine inference and must be modelled, not absorbed into noise.**

- *TV* sign depends on disease class: positive for acute recovery, near-zero for stable chronic, negative for progressive conditions.
- Selection-driven enrolment at peak severity guarantees positive *TV* in early trial weeks regardless of disease class.
- The PATH statement identifies treatment-effect heterogeneity modelling as mandatory for precision-medicine inference.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The time-variant natural-history component reflects whatever trajectory the participant would have followed without treatment. For acute conditions this trends positive (recovery on intrinsic timescale); for chronic stable conditions it is near zero on the population mean but with substantial participant-level variance; for chronic progressive con-

481 conditions it trends negative; and for selection-driven enrolment processes  
482 it is positive in the early weeks of the trial regardless of underlying  
483 disease class, because of regression to the mean<sup>11,13</sup>. The PATH  
484 statement<sup>14,15</sup> highlights that this heterogeneity is itself the central ob-  
485 ject of precision-medicine inference: heterogeneity of treatment effect  
486 must be modelled, not absorbed into noise, if the precision-medicine  
487 question is to be answered.

## 488 1.5 What this paper contributes

489 [15]

490 **The paper makes four contributions spanning formalisation,**  
491 **identifiability mapping, tractable methodology, and quantifi-**  
492 **cation of the inferential cost of lumping.**

- 493 • Sections 2–3 formalise the decomposition; section 4 maps it to  
494 design contrasts; section 5 develops tractable analysis methods.
- 495 • Sections 6–7 provide worked examples and a Monte Carlo simu-  
496 lation study; section 8 applies the framework to biomarker vali-  
497 dation.
- 498 • A companion pedagogical document provides extended worked  
499 examples for first-time readers.

500 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
501 *tempor incididunt ut labore et dolore magna aliqua.*

502 We make four contributions, each addressed in a section below. We  
503 begin in section 2 with the formal model and notation. Section  
504 3 develops each component in turn, with biological motivation,  
505 parametric form, and identifiability conditions. Section 4 maps the  
506 design-contrast structure of the standard hybrid open-label-plus-  
507 blinded-discontinuation design onto the identifiability requirements of  
508 the decomposition. Section 5 develops the analysis-side methodology,  
509 including the linear-mixed-effects approximations and the phase-by-  
510 treatment indicator extension that retains BR-PB sensitivity at low  
511 parametric cost. Section 6 presents the worked examples, and section  
512 7 presents the simulation study. Section 8 discusses applications to  
513 predictive-biomarker validation. The companion pedagogical docu-  
514 ment at docs/24-component-decomposition-pedagogy.md provides  
515 an extended worked-example treatment for readers approaching the  
516 material for the first time.

## 517 2 Notation and the formal model

518 [16]

519 **This section fixes compact notation and states the formal**

520 **model; a fuller worked-example treatment is in the compan-**  
521 **ion pedagogical document.**

- 522 • Definitions are precise enough to support the identifiability and  
523 analysis-side arguments in later sections.
- 524 • The presentation is kept compact; extended derivations are de-  
525 ferred to the companion document.
- 526 • The companion pedagogical document provides numerical worked  
527 examples at each formal step.

528 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
529 *tempor incididunt ut labore et dolore magna aliqua.*

530 In this section we shall fix notation and state the formal model.  
531 The three components are defined precisely enough to make  
532 the identifiability and analysis-side arguments in later sections  
533 rigorous, but we have endeavored to keep the presentation com-  
534 pact. A fuller treatment, with worked numerical examples at  
535 each step, is available in the companion pedagogical document at  
536 docs/24-component-decomposition-pedagogy.md.

537 [17] where  $Y_{it}$  is the observed symptom score,  $BL_i$  is participant  $i$ 's  
538 baseline score, the three components  $BR$ ,  $PB$ ,  $TV$  are non-negative  
539 reductions in symptom severity attributable to the three causal mech-  
540 anisms identified in section 1, and  $\varepsilon_{it}$  is observation-level noise. The  
541 sign convention is that an increase in any component reduces  $Y$ ; all  
542 three components are zero at  $t = 0$  by construction (no exposure yet  
543 to drug, no belief activation, no elapsed natural history) and grow over  
544 the duration of the trial.

545 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
546 *tempor incididunt ut labore et dolore magna aliqua.*

547 For participant  $i$  at trial timepoint  $t$ :

$$Y_{it} = BL_i - [BR_{it} + PB_{it} + TV_{it}] + \varepsilon_{it},$$

548 [18]

549 **Each component is driven by its own time-on-state variable,**  
550 **making the components structurally distinct even when they**  
551 **overlap in calendar time.**

- 552 •  $BR_{it}$  accumulates with time-on-drug;  $PB_{it}$  with time-on-belief  
553 scaled by expectancy  $\eta(\text{phase})$ ;  $TV_{it}$  with calendar time from  
554 trial entry.
- 555 • The distinct time indices are the structural source of identifiabil-  
556 ity from phase-contrast data.

- All components are zero at  $t = 0$  by construction and grow over the trial duration.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Each component is a participant-specific function of its own time-on-state variable.  $BR_{it}$  depends on time-on-drug, the cumulative exposure to active medication since drug initiation.  $PB_{it}$  depends on time-on-belief, modulated by a phase-specific expectancy factor  $\eta(\text{phase})$  that captures the patient's confidence that they are receiving active treatment.  $TV_{it}$  depends on time-since-trial-entry, calendar time elapsed regardless of treatment status.

[19] rescaled so that  $C(0) = 0$  and  $C(t) \rightarrow m_C(1 - e^{-d_C})$  as  $t \rightarrow \infty$ , for  $C \in \{BR, PB, TV\}$ . The three Gompertz parameters  $(m_C, d_C, r_C)$  are participant-specific random effects drawn from population distributions, and the participant-level heterogeneity in these distributions is the principal source of between-person response variability that the framework is designed to characterise.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Each component follows a modified Gompertz curve:

$$C(t) = m_C \exp(-d_C \exp(-r_C t)) - m_C \exp(-d_C),$$

[20]

**The residual term has AR(1) structure and the three components are permitted to co-vary through cross-component covariances.**

- Within-participant residuals follow  $\text{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$ .
- Within-time and across-time cross-component covariances permit participant-level heterogeneity in one component to co-vary with the others.
- This structure accommodates biologically plausible correlations, such as high- $BR$  patients also tending toward high- $PB$ .

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

$\varepsilon_{it}$  is a within-participant residual term with AR(1) serial-correlation structure  $\text{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$ . The three components are permitted to be jointly correlated through within-time and across-time cross-component covariances  $c_{cf1t}$  and  $c_{cfct}$  respectively, so that participant-level heterogeneity in any one component can co-vary with heterogeneity in the others.

Table 1: Principal psychometric and pharmacological terms used in the three-component decomposition.

| Term                 | Meaning                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR                   | Biological response: pharmacological reduction in symptom score attributable to active drug.                                                            |
| PB                   | Placebo-belief response: reduction attributable to the patient’s belief that they are receiving treatment.                                              |
| TV                   | Time-variant component: reduction (or increase) attributable to the natural-history trajectory of the disease.                                          |
| $\eta(\text{phase})$ | Expectancy multiplier for <i>PB</i> : 1 in open-label on-drug, $\approx 0.5$ in blinded discontinuation, between 0 and 1 in open-label discontinuation. |
| $BL_i$               | Participant $i$ ’s baseline (pre-treatment) symptom score.                                                                                              |
| $D_{it}$             | Continuous-decay drug indicator on the analysis side: 1 on-drug, $\exp(-\lambda t_{sd,it})$ off-drug.                                                   |
| $t_{sd,it}$          | Time since drug discontinuation (off-drug only).                                                                                                        |
| $\lambda$            | Carryover decay rate, with half-life $t_{1/2} = \ln 2 / \lambda$ .                                                                                      |

The principal psychometric and pharmacological terms used below are summarised in Table 1.

### 3 The three components

[21]

**This section presents each component with its biological motivation, Gompertz parameterisation, and identifiability conditions to establish the model as causally interpretable.**

- The three components are presented in turn: *BR* (biological response), *PB* (placebo-belief), *TV* (natural history).
- Each component is grounded in independent biological literature before its parametric form is stated.
- The section’s purpose is to demonstrate that the model sum reflects causal structure, not mathematical convenience.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

In this section we shall present the three components in turn, each with its biological motivation, the modified Gompertz parameterisation, and the design contrasts that identify it. The treatment is organised so that the reader can see, by the end of this section, that the formal model is a sum of three causally interpretable trajectories rather than a mathematical convenience.

#### 3.1 Biological response (BR)

[22]

***BR* is the pharmacological target of regulatory approval and biomarker stratification, independent of belief or natural history.**

- $BR$  arises from chemical action on the body and is zero for inert pills regardless of patient expectation.
- It is the estimand that approval decisions, dose selection, and biomarker-guided stratification require.
- Separating  $BR$  from  $PB$  and  $TV$  is the central precision-medicine problem the framework addresses.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

$BR$  is the physiological reduction in symptom score attributable to the active medication's chemical action on the body. It is independent of belief, suggestion, and natural history: a participant who somehow received active drug without knowing it would still develop a measurable  $BR$ , and a participant who took an inert sugar pill while believing it was active drug would have  $BR = 0$  no matter how strong their expectation. Methodologically,  $BR$  is the quantity that drug regulators and prescribers want: approval decisions, dose selection, and biomarker-guided patient stratification are all framed around  $BR$ .

[23]

**PK/PD considerations produce a saturating  $BR$  trajectory with a slow initial rise, a peak-growth phase, and a sharp drop to zero on discontinuation.**

- Plasma steady state is reached within a few half-lives; pharmacodynamic integration (e.g. receptor adaptation) adds additional lag.
- For prazosin-PTSD, Raskind et al. (2013) calibrate the onset to several weeks of sustained exposure.
- The saturating shape motivates the Gompertz over linear or simple exponential alternatives.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The qualitative shape of the  $BR$  trajectory follows from pharmacokinetic and pharmacodynamic considerations. Drug concentration in plasma reaches steady state within a small multiple of the elimination half-life; brain concentration follows plasma with a short delay; the downstream pharmacodynamic response (reduction in symptom-relevant neural activity, in the case of prazosin's effect on noradrenergic tone during REM sleep) typically requires several days to weeks of sustained exposure to integrate fully<sup>44</sup>. The result is a saturating curve: slow rise in the first week, steepest growth in the second to third week, plateau as receptor adaptation completes, and a decay back toward zero on discontinuation governed by the carryover half-life  $t_{1/2} = \ln 2/\lambda$  rather than an instantaneous drop. The rate of this

off-drug decay is the object of the carryover machinery used throughout this project; only in the limit of a short half-life relative to the measurement spacing does it approximate the sharp step assumed in the worked example of section 6.

[24]

**The three Gompertz parameters ( $m_{BR}, r_{BR}, d_{BR}$ ) capture ceiling, onset rate, and climb shape respectively, calibrated to prazosin-PTSD reference values.**

- $m_{BR}$  is the biological ceiling (maximum reduction at infinite exposure);  $r_{BR}$  the onset rate;  $d_{BR}$  the displacement shaping the early climb.
- Prazosin-PTSD calibration yields  $m_{BR} \approx 11$ ,  $r_{BR} \approx 0.42/\text{week}$ ,  $d_{BR} \approx 5$ .
- These are participant-specific random effects whose population distributions encode between-person response heterogeneity.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The modified Gompertz form  $BR(t) = m_{BR} \exp(-d_{BR} \exp(-r_{BR}t)) - m_{BR} \exp(-d_{BR})$  captures this profile compactly. The asymptote  $m_{BR}(1 - e^{-d_{BR}})$  is the biological ceiling, the largest reduction the drug can produce at infinite exposure; the scale parameter  $m_{BR}$  equals that ceiling up to the factor  $(1 - e^{-d_{BR}})$ , which for the calibrated  $d_{BR} \approx 5$  exceeds 0.99. The onset rate  $r_{BR}$  controls how rapidly the participant approaches the ceiling: high  $r_{BR}$  means a fast onset (effect plateaus in one to two weeks), low  $r_{BR}$  a slow climb (six to eight weeks). The displacement  $d_{BR}$  shapes the climb: high  $d_{BR}$  gives an early-onset curve that rises sharply at first, low  $d_{BR}$  a more gradual climb. For PTSD prazosin, calibrated values from the<sup>34</sup> reference dataset are  $m_{BR} \approx 11$  (eleven points of nightmare-score reduction at saturation),  $r_{BR} \approx 0.42$  per week (half-maximum at roughly four to five weeks of sustained exposure), and  $d_{BR} \approx 5$  (moderately steep early climb).

[25]

**The Gompertz is preferred over linear and simple-exponential alternatives because it captures the asymmetric slow-start saturation profile of chronic-condition pharmacodynamics.**

- Linear  $BR = \beta t$  is biologically implausible past the first few weeks due to unbounded growth.
- Exponential  $BR = m(1 - e^{-rt})$  lacks the slow-start phase from receptor adaptation and gene-expression changes.
- The Gompertz interpolates between both limits and its three-

parameter form avoids overfitting; sensitivity analysis is reported separately in 07-gompertz-evaluation.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Two alternative families that do not capture the asymmetric saturation profile are worth noting for contrast. A linear function  $BR = \beta t$  climbs without bound and is biologically implausible past a few weeks. An exponential decay  $BR = m_{BR}(1 - \exp(-rt))$  is acceptable for fast-onset drugs but lacks the slow-start phase that characterises most chronic-condition responses, where receptor adaptation, feedback regulation, and gene-expression changes produce a measurable lag. The Gompertz interpolates between exponential decay (high  $d$  limit) and linear-in- $t$  growth (low  $d$  limit), and its three-parameter form is flexible enough to fit a wide range of pharmacodynamic profiles without overfitting. The Gompertz is a deliberate choice over the two parametric forms most familiar from pharmacokinetic-pharmacodynamic modelling. The sigmoid  $E_{\max}$  (Hill) model<sup>45</sup> describes the dependence of effect on concentration (a dose-response curve), whereas  $BR(t)$  here is a time-course of accumulating effect at fixed dosing; and the indirect-response (turnover) models that do describe effect time-courses are specified through differential equations on a hidden response variable rather than in closed form. The modified Gompertz retains the slow-start, saturating shape those models produce while remaining a closed-form, three-parameter mean function that is tractable inside a mixed-effects likelihood. Doc 25 of the project documentation provides the full historical and biological context for the family choice and quantifies the sensitivity of pmsimstats inference to alternative families; that work is reported separately as 07-gompertz-evaluation. The headline result of that evaluation is reassuring for the present framework: in a 1{,}000-replicate-per-cell simulation that generated data under the logistic, hyperbolic-tangent, and piecewise-linear breakpoint families and analysed them under the standard linear-mixed model, the trajectory family was essentially irrelevant to the power of the biomarker-treatment interaction test (the family-to-family spread fell within two Monte Carlo standard errors of sampling noise) and to type I error (0.025 to 0.036 across null cells, with no family-specific inflation). The conclusions drawn here are therefore not artefacts of the Gompertz form, though the regimes the family cannot represent (cyclical or biphasic trajectories) remain genuine exclusions.

[26]

**$BR$  is identified by the blinded-discontinuation contrast, where drug removal collapses  $BR$  to zero while  $PB$  remains intact.**

- Within-participant on-drug versus off-drug contrasts are the primary identification mechanism.
- The blinded-discontinuation phase provides the cleanest contrast:  $BR$  drops,  $PB$  is preserved.
- This contrast does not require the analyst to model  $PB$  or  $TV$  explicitly to recover  $BR$ .

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The  $BR$  component is identified by within-participant on-drug versus off-drug contrasts, formalised in section 4 below. The most direct contrast is the blinded-discontinuation phase: when a participant is silently switched to placebo while still believing they are on active drug,  $BR$  decays toward zero as the drug clears while  $PB$  remains roughly intact, and the difference between on-drug and off-drug response in that window is a relatively clean estimate of  $BR$  that does not require the analyst to model  $PB$  or  $TV$  explicitly. The decay is not instantaneous: with the carryover half-life  $t_{1/2} = \ln 2 / \lambda$  of the analysis-side decay term, residual  $BR$  persists into the early off-drug observations, so the contrast is clean only once the discontinuation window is long relative to  $t_{1/2}$ , and the drug state is best carried by the continuous-decay indicator  $D_{it}$  of Table 1 rather than a hard on/off indicator.

### 3.2 Placebo-belief response (PB)

[27]

**$PB$  is the strict placebo response: improvement that persists when the active drug is silently replaced, provided the patient’s belief is unchanged.**

- $PB$  is belief-dependent, not drug-dependent; it would survive a silent switch to placebo under sustained belief.
- It is distinct from  $BR$  in causal origin:  $BR$  requires chemical action;  $PB$  requires only belief.
- The definition is strict and operational, not vague or colloquial.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

$PB$  is the reduction in symptom score that arises from the patient’s belief that they are receiving active treatment, independent of what they are actually receiving. That is, it is the placebo response in its strict sense: the part of the improvement that would persist if the active drug were silently replaced with an inert pill, provided the patient continued to believe they were on the active drug.

[28]

***PB* reflects neurobiologically real mechanisms triggered by belief, not pharmacology; neuroimaging and genetic evidence confirm it is a structured biological phenomenon.**

- Belief-driven endogenous opioid/dopamine release, HPA-axis modulation, and top-down attentional shifts are all documented mechanisms.
- Wager et al. neurologic pain signature dissociates pharmacological from expectancy contributions via fMRI.
- COMT genotype predicts placebo responsiveness, confirming biological structure rather than measurement artefact.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

It is worth being precise about what *PB* is and is not. *PB* is not ‘fake’ improvement; it is a phenomenon clinicians knowingly exploit, a national survey of US internists and rheumatologists finding roughly half report regular use of placebo treatments<sup>46</sup>. As the introduction sets out, the placebo response is mediated by well-characterised neurobiological pathways<sup>38–41,47</sup>: belief-driven release of endogenous opioids and dopamine, anticipatory hypothalamic-pituitary-adrenal regulation, and top-down attentional shifts, dissociated from pharmacology by the<sup>42</sup> neurologic pain signature<sup>43</sup> and shown to have genetic correlates<sup>9,10</sup>. What makes these mechanisms ‘placebo’ rather than ‘drug’ is that they are triggered by belief rather than by pharmacology.

[29]

***PB* is identifiable from *BR* only when belief and treatment allocation differ, as in the blinded-discontinuation phase.**

- The blinded-discontinuation phase is the canonical contrast: patients cannot tell whether they remain on active drug.
- Belief drops to a partial level ( $\eta \approx 0.5$ ) because the patient’s expectation is hedged, not absent.
- Designs without a phase that decouples belief from treatment cannot separately identify *PB* from *BR*.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

For trial design, the relevant point is that *PB* is separately identifiable from *BR* if and only if the trial includes a phase in which the patient’s belief about treatment differs from the actual treatment they are receiving. The blinded-discontinuation phase is the canonical such phase: the patient is told that, at some unannounced point during the window, they may be silently switched to placebo, and they cannot reliably tell whether they are still receiving active drug. Belief in

830 treatment correspondingly drops to a partial level (multiplier roughly  
831 0.5) because the patient's expectation is hedged.

832 [30]

833 **The  $\eta = 0.5$  blinded-phase expectancy is a modelling assumption**  
834 **operationalising response expectancy, not a measured**  
835 **quantity, and requires sensitivity analysis.**

- 836 • The construct is response expectancy in the sense of Kirsch  
837 (1985); to our knowledge the residual expectancy under blinded  
838 discontinuation has not been measured directly.
- 839 • Open-label placebo trials confirm clinically meaningful *PB*-only  
840 responses even under explicit disclosure, bounding the residual  
841 response away from zero.
- 842 • Sensitivity analyses varying  $\eta$  are mandatory before clinical con-  
843 clusions depend on this single parameter.

844 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
845 *tempor incididunt ut labore et dolore magna aliqua.*

846 *PB* also follows a Gompertz curve, scaled by the expectancy factor:

$$PB(t \mid \text{phase}) = \eta(\text{phase}) \cdot m_{PB} \exp(-d_{PB} \exp(-r_{PB}t)) + (\text{offset to start at zero}).$$

847 Typical expectancy values are  $\eta = 1.0$  in open-label on-drug,  $\eta \approx 0.5$  in  
848 blinded discontinuation, and  $\eta$  between 0 and 0.5 in open-label placebo  
849 (the residual expectation a patient retains when told the previous ac-  
850 tive phase was beneficial). The quantity  $\eta$  operationalises response  
851 expectancy in the sense of<sup>48</sup>, the patient's anticipation of benefit that  
852 the expectancy literature treats as a determinant of experienced symp-  
853 tom change. The choice of  $\eta = 0.5$  for the blinded phase is a modelling  
854 assumption rather than a calibrated quantity: to our knowledge the  
855 residual expectancy a patient carries when uncertain about allocation  
856 has not been measured directly as a fraction of the open-label level,  
857 and we adopt 0.5 as a deliberately central placeholder. The avail-  
858 able indirect evidence is nonetheless that the residual response is sub-  
859 stantial. Open-label placebo trials<sup>35–37</sup> demonstrate that a clinically  
860 meaningful response persists even when patients are explicitly told  
861 they are receiving an inert intervention, which bounds the residual  
862 belief-driven response away from zero. Because the value of  $\eta$  is not  
863 empirically pinned, sensitivity analyses varying it across its plausible  
864 range are mandatory before any conclusion is allowed to depend on  
865 this single number.

866 [31]

867 **Kaptchuk et al. (2008) provide the structural precedent; the**

868 **BR-PB-TV framework extends the same decomposition logic**  
869 **to the full treatment response.**

- 870 • Kaptchuk decomposed the placebo response itself into observa-  
871 tion, ritual, and relationship components empirically.
- 872 • The present framework applies analogous decomposition logic at  
873 the level of drug, placebo, and natural history.
- 874 • Both frameworks reject the 'expectancy lump' model in favour of  
875 identifiable additive parts.

876 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
877 *tempor incididunt ut labore et dolore magna aliqua.*

878 The decomposition framework as proposed here has a structural ana-  
879 logue in the placebo-research literature.<sup>5</sup> decomposed the placebo re-  
880 sponse itself into observation, ritual, and patient-practitioner relation-  
881 ship components in a randomised trial of acupuncture-style placebo for  
882 irritable bowel syndrome, demonstrating empirically that the placebo  
883 response is not a single 'expectancy' lump but a sum of identifiable  
884 parts. The present BR-PB-TV framework is analogous in spirit but  
885 operates at the level of the full treatment response (drug, placebo,  
886 natural history) rather than within the placebo response alone.

887 [32]

888 **PB variance scales with  $\eta$ , not just its mean; ignoring this**  
889 **heteroscedastic structure miscalibrates standard errors and**  
890 **downstream biomarker tests.**

- 891 • Phase-dependent variance follows  $SD(PB \mid \text{phase}) = \eta(\text{phase}) \cdot$   
892  $\sigma_{PB}^{\text{baseline}}$ .
- 893 • Open-label phases have high between-patient  $PB$  variance;  
894 blinded phases have low variance.
- 895 • Ignoring heteroscedasticity deflates standard errors in open-label  
896 and inflates them in blinded phases, with consequences for CI  
897 coverage and biomarker test calibration.

898 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
899 *tempor incididunt ut labore et dolore magna aliqua.*

900 A subtle property of  $PB$  is that its variance also scales with  $\eta$ ,  
901 not just its mean. When  $\eta$  is high, there is substantial variability  
902 across patients in how strongly the placebo response is expressed;  
903 high-suggestibility patients show large  $PB$ , low-suggestibility pa-  
904 tients show essentially none. When  $\eta$  is low (blinded), the placebo  
905 response is uniformly attenuated and the patient-to-patient vari-  
906 ance shrinks accordingly. The model therefore parameterises  
907  $SD(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$ . Ignoring this heteroscedastic  
908 structure deflates standard errors in the open-label phase and inflates  
909 them in the blinded phase, with downstream consequences for

910 confidence-interval coverage and for the calibration of any biomarker  
911 test built on top of the analysis.

### 912 3.2.1 Additivity, subadditivity, and the BR-PB interaction

913 [33]

914 **Additivity is a modelling assumption, not biological fact; the**  
915 **section examines whether a  $BR \times PB$  interaction term is war-**  
916 **ranted.**

- 917 • The formal model writes total response as a sum; this section  
918 asks whether the  $BR$ - $PB$  interaction is zero.
- 919 • The question has inferential consequences for biomarker valida-  
920 tion and for what the phase-augmented analysis captures.
- 921 • The treatment is preparatory for the generalised interaction  
922 model introduced later in this section.

923 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
924 *tempor incididunt ut labore et dolore magna aliqua.*

925 In this section we shall broach a question that the formal model leaves  
926 open: whether the three components contribute additively to symptom  
927 reduction, or whether there is a meaningful interaction between  $BR$   
928 and  $PB$  that the additive specification misses.

929 [34]

930 **The additive model may be wrong: published literature doc-**  
931 **uments subadditivity between pharmacological and placebo-**  
932 **belief effects in some clinical contexts.**

- 933 • Subadditivity means the combined  $BR + PB$  effect is less than  
934 the sum of each component measured separately.
- 935 • Evidence from Kaptchuk et al. and Benedetti et al. shows the  
936 phenomenon is context-dependent, not universal.
- 937 • Formal test:  $\partial Y / \partial (BR \cdot PB) > 0$  in a saturated mean-structure  
938 model.

939 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
940 *tempor incididunt ut labore et dolore magna aliqua.*

941 The decomposition as written in section 2 treats the three components  
942 as additive contributors to symptom reduction:

$$Y_{it} = BL_i - (BR_{it} + PB_{it} + TV_{it}) + \varepsilon_{it}.$$

943 The additivity is a modelling assumption, not a fact about biology,  
944 and it is worth stating that the assumption may be wrong in direc-  
945 tions the framework can absorb and in directions it cannot. Additiv-

ity is also scale-dependent in the sense emphasised by<sup>49</sup>: components that combine non-additively on the raw symptom scale may combine additively on a transformed (for example logarithmic) scale, so an apparent  $BR \times PB$  interaction can be partly an artefact of the chosen response metric rather than a biological fact. That belief and pharmacology interact rather than act in isolation is demonstrated directly by<sup>50</sup>, who showed with neuroimaging that positive treatment expectation roughly doubled the analgesic benefit of the opioid remifentanyl while negative expectation abolished it, so the two channels are not in general separable by simple addition. More specifically, the published literature on placebo-drug interactions has accumulated evidence that the combined effect of receiving an active drug and believing one is receiving an active drug is, in some clinical contexts, less than the sum of the two components measured separately<sup>5,38</sup>. The phenomenon is typically termed subadditivity:  $\partial Y / \partial (BR \cdot PB) > 0$  when both components are present and their interaction term is included in a saturated mean-structure model.

[35]

### Three candidate mechanisms for subadditivity are shared neural substrate, scale ceiling effects, and belief-dynamics adaptation.

- Shared pain-modulation circuits produce saturating occupancy when both  $BR$  and  $PB$  activate them simultaneously.
- Scale ceiling effects prevent registration of additional symptom reduction once  $BR + PB$  saturates the measurement range.
- Belief-dynamics adaptation: unexpectedly large  $BR$  can recalibrate patient expectation downward, attenuating ongoing  $PB$ .

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Three mechanisms have been proposed in the placebo-research literature to explain subadditivity. First, a shared neural substrate: if the descending pain-modulation circuits that mediate placebo analgesia overlap with the circuits modulated by the active drug, simultaneous activation by both mechanisms produces less marginal benefit than activation by either alone, in a saturating-occupancy sense. Second, ceiling effects on the symptom scale: a clinical scale that caps at zero (no symptoms) cannot register additional reduction once a participant has reached that cap, so any participant whose  $BR + PB$  already saturates the scale will register no further reduction even if  $BR$  or  $PB$  would, in isolation, predict more. Third, belief-dynamics adaptation: the patient's expectation may adapt downward when the active drug produces a strong response that exceeds the expected effect, reducing the ongoing  $PB$  contribution; equivalently, unexpectedly large  $BR$

989 may attenuate  $PB$  via expectation recalibration.

990 [36] where  $\gamma < 0$  encodes subadditivity,  $\gamma = 0$  recovers the additive  
991 form, and  $\gamma > 0$  encodes superadditivity, is the natural extension.  
992 The interaction coefficient  $\gamma$  is identifiable from the same trial-design  
993 contrasts that identify the marginal components: open-label phases  
994 supply  $BR \cdot PB$  products at high  $\eta$ , blinded phases supply  $BR \cdot PB$   
995 products at attenuated  $\eta$ , and the  $\gamma$  coefficient is estimated from the  
996 difference between the two regimes once the marginal components are  
997 accounted for.

998 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
999 *tempor incididunt ut labore et dolore magna aliqua.*

1000 Whichever mechanism dominates in a given clinical context, the subad-  
1001 ditivity question is empirical and design-dependent. Within the BR-  
1002 PB-TV framework, subadditivity manifests as a non-zero coefficient  
1003 on a  $BR \times PB$  interaction term in the conditional-mean specification.  
1004 The additive form treats this interaction as zero by construction; a  
1005 generalised form

$$Y_{it} = \text{BL}_i - [BR_{it} + PB_{it} + TV_{it} + \gamma \cdot BR_{it} \cdot PB_{it}] + \varepsilon_{it},$$

1006 [37]

1007 **Subadditivity biases biomarker predictive accuracy estimates**  
1008 **and is partly absorbed by the phase-augmented analysis even**  
1009 **without explicit  $\gamma$  estimation.**

- 1010 • Under subadditivity, the biomarker-by- $BR$  association is partly  
1011 cancelled, inflating apparent predictive accuracy in the additive  
1012 model.
- 1013 • Phase designs attenuating  $PB$  (blinded, crossover) push the effec-  
1014 tive biomarker-treatment estimate closer to the true biomarker-  
1015 by- $BR$  value.
- 1016 • The phase-augmented analysis in section 5 absorbs subadditivity  
1017 as a side benefit.

1018 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
1019 *tempor incididunt ut labore et dolore magna aliqua.*

1020 For trial design, two consequences follow. First, a biomarker that  
1021 predicts  $BR$  in an additive-decomposition analysis may overstate or  
1022 understate predictive accuracy if the true mechanism is subadditive:  
1023 under subadditivity, the biomarker-driven increase in  $BR$  is partly  
1024 cancelled by the  $\gamma \cdot BR \cdot PB$  interaction, and the marginal biomarker-  
1025 treatment association is smaller than the biomarker-by- $BR$  associa-  
1026 tion. Second, trial designs that attenuate  $PB$  in some phases (blinded

discontinuation, crossover) implicitly down-weight the subadditivity contribution to the response in those phases, which means that the effective biomarker-treatment association estimated from those phases is closer to the biomarker-by- $BR$  association than is the open-label estimate. The phase-augmented analysis recommended in section 5 below therefore has the side benefit of partly absorbing subadditivity, even when  $\gamma$  is not estimated explicitly.

[38]

**A forthcoming simulation study will quantify how bias in the biomarker-treatment interaction scales with  $\gamma$ ; the pre-registration predicts linear scaling.**

- Bias in  $\hat{\beta}_{bm:D}$  is expected to scale linearly with  $\gamma$  at small  $|\gamma|$ .
- Under subadditivity, bias sign is opposite to the true biomarker-by- $BR$  effect; under superadditivity it is aligned.
- Empirical quantification is deferred to the production simulation programme in section 7.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

A simulation study quantifying the magnitude of bias in the biomarker-treatment interaction estimate when the additive analysis is fitted to data generated under non-zero  $\gamma$  is forthcoming as part of the simulation programme described in section 7. The pre-registration is that bias scales linearly with  $\gamma$  at small  $|\gamma|$ , with the sign opposite to the true biomarker-by- $BR$  effect under subadditivity and aligned with it under superadditivity.

[39]

**Non-zero  $\hat{\gamma}$  does not establish true subadditive interaction; latent-class structure with class-correlated components produces the same marginal pattern.**

- If class membership correlates with both  $BR$  and  $PB$  strength, the marginal covariance of  $BR \cdot PB$  mimics a direct interaction term.
- This confounding is invisible from the lumped-analysis perspective but is identifiable in a class-aware analysis.
- Apparent subadditivity can arise from latent population heterogeneity with no direct  $BR$ - $PB$  interaction in the class-conditional DGP.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

A subtle but inferentially important point merits attention here: a non-zero  $\hat{\gamma}$  recovered from an additive-extension analysis does not,

on its own, establish that the underlying mechanism is subadditive interaction in the strict sense. The same marginal pattern can arise from latent population structure that the analyst is not modelling. If the trial population is a mixture of treatment-responders and non-responders (in the latent-class sense developed at length in [analysis/report/03-latent-class-mixture-application/](#)), and class membership correlates with both  $BR$  strength and  $PB$  strength (responder-class participants have both higher  $BR$  and higher  $PB$  on average), then the marginal expected value of  $BR \cdot PB$  over the unobserved class label has a non-zero covariance term whose magnitude depends on the gating function  $\pi(B)$ . From the lumped-analysis perspective this looks exactly like a  $\gamma$ -weighted  $BR \times PB$  interaction even though no such direct interaction was specified in the class-conditional DGP. That is, the latent-class structure is one mechanism that produces apparent subadditivity in a one-component analysis, regardless of whether the underlying biology has a true direct  $BR$ - $PB$  interaction.

[40]

**Distinguishing true subadditivity from latent-class confounding requires either a class-aware analysis or the phase contrasts the BR-PB-TV framework provides.**

- The two mechanisms produce qualitatively similar marginal effects but are inferentially distinct.
- $\hat{\gamma}$  should be reported alongside a class-aware sensitivity analysis, not as standalone evidence for mechanistic subadditivity.
- The latent-class paper’s `pb_class_correlation` simulation axis is the direct test; cross-paper synthesis is the cleanest disentanglement.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The two mechanisms, namely true subadditive interaction and latent-class structure with class-correlated components, produce qualitatively similar marginal effects but are inferentially distinct. Distinguishing them requires either a class-aware analysis that explicitly models the latent population structure (as the latent-class paper develops at length), or a phase contrast that separates the mechanisms by exploiting the differential expectancy modulation of  $PB$  that the BR-PB-TV framework provides. Under the additive-extension analysis fitted to mixture-DGP data with class-correlated components,  $\hat{\gamma}$  is identified up to confounding with the class structure, and the analyst should report  $\hat{\gamma}$  alongside a class-aware sensitivity analysis rather than as standalone evidence for mechanistic subadditivity. The Study 1 simulation in the latent-class paper includes a `pb_class_correlation` axis that varies

$\text{Cor}(BR, PB)$  across class labels; that axis is the direct simulation-side test of the latent-class generation of apparent subadditivity, and the cross-paper synthesis between the two papers is the cleanest way to disentangle the two mechanisms.

### 3.3 Time-variant natural-history component (TV)

[41]

**$TV$  is the untreated counterfactual trajectory; it is participant-specific, structured, and can be positive, negative, or near-zero depending on disease class.**

- $TV$  is the symptom change that would have occurred without treatment, integrated over the trial duration.
- It varies in sign across disease classes and across patients within a class.
- It is independent of treatment status and belief, distinguishing it structurally from  $PB$ .

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

$TV$  is the reduction in symptom score that would have happened even if the patient had received no treatment at all. It is the participant's natural-history trajectory, integrated over the duration of the trial. For some participants  $TV$  is positive (symptoms naturally improve over the trial because of life events, therapy, time-in-condition, or regression to the mean); for others  $TV$  is negative (symptoms naturally worsen because of trauma anniversaries, accumulating stress, or progressive underlying disease).

[42]

**$TV$  is a structured participant-specific signal, not noise; its distinguishing feature from  $PB$  is independence from treatment status and belief.**

- Unlike  $\varepsilon_{it}$ ,  $TV$  is participant-level (slow trajectory) and requires explicit modelling, not absorption into noise.
- Off-drug baselines and follow-up windows are the primary data sources for estimating  $TV$ .
- $TV$  is modulated by neither treatment assignment nor patient belief, whereas  $PB$  is modulated by both.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Like  $PB$ ,  $TV$  is not noise. It is a structured participant-specific signal that the model can in principle estimate from data, given sufficient repeated measurements and a trial design that includes a long enough

1151 off-drug baseline or follow-up window. What distinguishes  $TV$  from  
1152  $PB$  is its origin:  $TV$  is independent of treatment status and of belief  
1153 about treatment, whereas  $PB$  is modulated by both.

### 1154 3.3.1 When does natural history actually exist?

1155 [43]

1156 **The shape of  $TV$  depends on disease class, trial timescale,**  
1157 **and selection process; none of these can be assumed and all**  
1158 **must be recovered empirically.**

- 1159 • The assumption that subjects improve over time is true for some  
1160 conditions and systematically wrong for others.
- 1161 • Disease class, trial timescale, and enrolment selection jointly de-  
1162 termine the sign and magnitude of  $TV$ .
- 1163 • Recovering the empirical shape of  $TV$  is one of the decomposi-  
1164 tion's primary objectives.

1165 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
1166 *tempor incididunt ut labore et dolore magna aliqua.*

1167 A common simplifying assumption in trial analysis is that subjects will  
1168 improve over time, with or without treatment. This is true for some  
1169 conditions and problematic for others. The shape of  $TV$  depends  
1170 on the underlying disease class, the trial timescale, and the selection  
1171 process by which participants entered the trial. None of these can be  
1172 assumed without thought, and the empirical shape of  $TV$  is one of the  
1173 things the decomposition is designed to recover.

1174 [44]

1175  **$TV$  sign varies systematically by disease class: positive for**  
1176 **acute recovery, near-zero for stable chronic, negative for pro-**  
1177 **gressive, and structured for cyclical conditions.**

- 1178 • Acute conditions (post-operative pain, acute back pain) resolve  
1179 intrinsically; substantial  $TV$ -improvement appears in the placebo  
1180 arm.
- 1181 • Progressive conditions (Alzheimer's, ALS) require explicit  
1182 negative-trajectory modelling; apparent treatment effect is  
1183 slowing of progression.
- 1184 • Cyclical/triggered conditions (PTSD with trauma anniversaries,  
1185 relapsing-remitting MS) require covariates beyond the Gompertz  
1186 form.

1187 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
1188 *tempor incididunt ut labore et dolore magna aliqua.*

1189 Acute conditions tend toward  $TV$ -positive: a sprained ankle, a post-  
1190 operative pain syndrome, an episode of acute back pain all resolve on a

timescale of days to weeks even without intervention, and a trial of an analgesic in such a setting will see substantial  $TV$ -improvement in the placebo arm. Chronic stable conditions (treated hypertension, controlled chronic asthma, long-standing PTSD without active triggers) tend toward  $TV$ -zero on the population mean but with substantial individual variance. Chronic progressive conditions (Alzheimer's disease, ALS, late-stage congestive heart failure, many cancers) tend toward  $TV$ -negative: a trial of a neuroprotective agent in early Alzheimer's must explicitly model the negative natural-history trajectory, since the apparent treatment effect is the slowing of progression, not absolute improvement. Cyclical and triggered conditions (PTSD with trauma anniversaries, seasonal-pattern major depression, relapsing-remitting multiple sclerosis) show structured  $TV$  that is neither monotone nor stationary, and may require explicit covariates beyond the Gompertz form.

[45]

**Selection-induced  $TV$  is real and rarely zero even for stable chronic conditions, because enrolment coincides with peak severity triggering help-seeking.**

- Regression to the mean alone contributes several points of apparent improvement in early trial weeks.
- Hrobjartsson et al. systematic reviews across 200+ trials show much of conventional placebo response is indistinguishable from natural-history drift.
- Kirsch (2008) and Locher (2015) demonstrate the clinical consequences in antidepressant and late-life depression trials.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Selection-induced  $TV$  is real and rarely zero, even when the underlying condition is stable. Patients enrol at moments of clinical engagement that often coincide with recent worsening, so regression to the mean alone produces several points of apparent  $TV$ -improvement in the early weeks of a trial. As the introduction documents, the systematic-review evidence of<sup>11</sup>, <sup>12</sup>, and<sup>13</sup> (the last covering 202 trials across 60 conditions) and the antidepressant analyses of<sup>6</sup> and<sup>7</sup> show that much of what is conventionally attributed to placebo response is indistinguishable from regression to the mean and natural-history drift.

[46]

**Trial participation itself induces  $TV$ -improvement through Hawthorne, structured-visit, and diary-completion effects, independent of pharmacology or placebo.**

- Hawthorne, structured-visit, and diary-completion effects each

1233 contribute symptom improvements attributable to participation,  
1234 not treatment.

- 1235 • These contributions are absorbed into  $TV$  in the decomposition,  
1236 not into  $BR$  or  $PB$ .
- 1237 • Ignoring these participation-induced effects inflates the apparent  
1238 pharmacological and placebo components.

1239 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
1240 *tempor incididunt ut labore et dolore magna aliqua.*

1241 Trial participation itself also induces  $TV$ : the Hawthorne effect, the  
1242 structured-visit effect, and the diary-completion effect appear in the  
1243 data as  $TV$ -improvement that is independent of any pharmacological  
1244 or placebo response.

### 1245 3.3.2 Can we assume no $TV$ effect?

1246 [47]

1247 **Assuming  $TV = 0$  is rarely safe in chronic psychiatric or pain**  
1248 **trials; the conservative default is to estimate  $TV$  from the**  
1249 **data.**

- 1250 • Zero- $TV$  assumption is valid only when all four conditions hold:  
1251 stable disease, short trial, non-severity-peaked recruitment, no  
1252 ancillary care.
- 1253 • For most chronic conditions at least one condition fails; the as-  
1254 sumption should be tested, not asserted.
- 1255 • Statistical efficiency lost by estimating  $TV$  is smaller than the  
1256 bias introduced by assuming it zero.

1257 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
1258 *tempor incididunt ut labore et dolore magna aliqua.*

1259 But how safe is it to assume  $TV$  is zero? The short answer is: rarely  
1260 safe, and never safe without empirical support. The longer answer is  
1261 that  $TV$  may be assumed zero only if the disease is in a stable chronic  
1262 phase, the trial is short relative to the disease timescale, the recruit-  
1263 ment process does not favour participants at peak severity, and the  
1264 trial protocol does not introduce ancillary care that itself improves  
1265 symptoms. For most chronic trials in psychiatric or pain conditions,  
1266 at least one of these conditions fails. The conservative default is to es-  
1267 timate  $TV$  from the data, even at the cost of some statistical efficiency,  
1268 rather than assume it is zero.

1269 [48]

1270 **The decomposition makes the zero- $TV$  assumption testable**  
1271 **and shows that lumping  $TV$  into residuals produces auto-**  
1272 **correlated, heteroscedastic residuals that bias  $BR$  and  $PB$**

standard errors.

- $\hat{m}_{TV}$  with a variance estimate from the model provides an empirical test of the zero- $TV$  assumption.
- Participant-level  $TV$  is structured and correlated with the on/off-drug schedule; it is not independent noise.
- Lumping  $TV$  into  $\varepsilon$  produces autocorrelated within-participant residuals and heteroscedastic cross-participant residuals.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The decomposition makes this assumption testable. A model that includes a  $TV$  component will return  $\hat{m}_{TV}$  with a variance estimate; if both are small relative to  $\hat{m}_{BR}$  and  $\hat{m}_{PB}$ , the analyst has empirical support for the simplifying assumption. If  $\hat{m}_{TV}$  is meaningfully large, the assumption was wrong, and the trial needs the  $TV$  component to produce unbiased inference about the others. Setting  $TV$  to zero and lumping it into  $\varepsilon$  is worse than estimating it: the participant-level  $TV$  is structured (each participant has their own slow trajectory) and not independent of the on-drug-versus-off-drug schedule, so lumping it into noise produces residuals that are autocorrelated within participants and heteroscedastic across them. The resulting standard errors on the  $BR$  and  $PB$  estimates are wrong in both directions: too small for participants whose  $TV$  is small, too large for participants whose  $TV$  is large. Estimating  $TV$  explicitly fixes both problems simultaneously.

### 3.3.3 Why a Gompertz, not a linear time covariate?

[49]

**A linear week covariate is insufficient for  $TV$  because it enforces linearity, ignores participant heterogeneity, and cannot distinguish  $TV$  from  $BR$  without the explicit decomposition.**

- A **week** covariate constrains each additional week to contribute a fixed natural-history change, misrepresenting curved or saturating disease trajectories.
- A single coefficient estimates a population-mean trajectory, ignoring participant-level heterogeneity in natural history.
- Without the decomposition and design contrasts, **week** absorbs both  $BR$  and  $TV$ , making them inseparable.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

A natural objection is that a model with **week** as a covariate already controls for time. But how adequate is such a control? Three reasons

suggest it is not enough. First, a `week` covariate enforces a linear trajectory, constraining each additional week to contribute a fixed amount of natural-history change; real disease trajectories curve, plateau, or accelerate, and the Gompertz form captures the S-shaped or saturating patterns linear time cannot. Second, a single `week` coefficient estimates a population-mean trajectory that ignores the participant-level heterogeneity in natural history. Third, *BR* and *TV* both evolve smoothly over the same trial timescale, and a `week` covariate cannot tell them apart from observational data alone; the design contrasts discussed in section 4 are what supply identifiability, and without the explicit decomposition the `week` coefficient absorbs both.

## 4 Identifiability and trial-design contrasts

[50]

**Identifiability is supplied by trial design; each phase of the hybrid design contributes a different combination of components to the observed trajectory.**

- The decomposition is mathematically specified but requires trial-design contrasts to be statistically identified.
- This section characterises the hybrid open-label-plus-blinded-discontinuation design as the primary identifiability vehicle.
- Each of the three phases supplies a distinct mixture of *BR*, *PB*, and *TV*.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The mathematical decomposition above is meaningless unless the data contain enough information to pin down each component separately. Trial design is what supplies that information. In this section we shall characterise the hybrid open-label-plus-blinded-discontinuation design, used as the worked example throughout this paper, and show how each of its three phases contributes a different combination of components to the observed trajectory.

| Phase                   | Drug?  | Belief? | Components present                                           |
|-------------------------|--------|---------|--------------------------------------------------------------|
| Open-label on drug      | Yes    | Full    | $BR + PB(\eta = 1) + TV$                                     |
| Blinded discontinuation | Hidden | Hedged  | $BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 0.5) + TV$ |
| Open-label crossover    | Either | Knows   | $BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 1) + TV$   |

Three phase-by-allocation contrasts isolate  $PB$ ,  $BR$ , and  $TV$  respectively; the LME model combines them but the inference is driven by design.

- Open-label versus blinded contrast yields approximately  $0.5 \cdot PB$ , isolating the belief component conditional on a known blinded-phase  $\eta$ .
- Blinded on-drug versus blinded placebo yields  $BR$  once drug carryover has elapsed (since  $PB$  and  $TV$  match across arms within the blinded phase).
- Off-drug phases yield  $TV$  plus residual off-drug  $PB$ , with no active  $BR$  contribution beyond carryover.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Three phase-by-allocation contrasts then identify the three components. The open-label versus blinded contrast (within drug) yields  $PB(\eta = 1) - PB(\eta = 0.5) \approx 0.5 \cdot PB$ , which isolates the belief component, but only conditional on a known blinded-phase expectancy  $\eta$ : from a two-phase design just the product  $\eta \cdot m_{PB}$  is identified, and separating  $\eta$  from  $m_{PB}$  requires the open-label placebo phase. The blinded on-drug versus blinded placebo contrast yields  $BR$  once drug carryover has elapsed, since  $PB$  and  $TV$  match across allocation arms within the blinded phase. All phases off drug yield  $TV$  plus any residual off-drug  $PB$ , with no active  $BR$  contribution beyond carryover and  $PB$  at the off-drug expectation level. The  $\mathbf{1}_{\text{on drug}}$  indicator in the phase table above is shorthand for the continuous-decay drug state  $D_{it}$  defined in Table 1.

This conditioning on  $\eta$  is the framework's principal identification limitation and deserves emphasis rather than a parenthetical. With a two-phase (open-label and blinded) design the data identify only the product  $\eta \cdot m_{PB}$ , so  $PB$  is recovered up to the assumed blinded-phase expectancy; the working value  $\eta \approx 0.5$  is a modelling assumption, not an estimate, and absent a belief or expectancy measurement it is not falsifiable from the outcome data alone. The identified set for  $m_{PB}$  is accordingly the interval  $\{\eta \widehat{m_{PB}} / \eta : \eta \in (0, 1]\}$ , which collapses to a point only when  $\eta$  is pinned down. Two design features close the gap: an open-label placebo phase, which supplies a second equation separating  $\eta$  from  $m_{PB}$ , and direct measurement of belief, as in the open-hidden and balanced-placebo designs<sup>51,52</sup> that manipulate or record the patient's treatment expectation. We therefore recommend that any trial relying on the decomposition for a  $PB$ -versus- $BR$  attribution either include such a phase or collect an expectancy measure, and that analyses report sensitivity of the attribution to  $\eta$  across its

plausible range.

[52]

**Designs lacking specific phases lose identifiability of the corresponding component; the decomposition machinery is only as useful as the design allows.**

- A pure open-label design cannot separate  $BR$  from  $PB$  because both are present at full strength throughout.
- A pure blinded design cannot estimate  $\eta = 1$   $PB$  because no phase achieves full-strength belief.
- Trial designers should verify each component has at least one phase in which it is uniquely present before committing to the design.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

A trial that lacks any of these phases loses the corresponding identifiability. A pure open-label design (no blinded discontinuation) cannot separate  $BR$  from  $PB$ , because both are present at full strength throughout. A pure blinded design (no open-label phase) cannot estimate  $\eta = 1$   $PB$ , because no phase has belief at full strength. The decomposition machinery is useful only to the extent that the design supplies identifiability, and trial designers should evaluate any proposed design by checking that each component has at least one phase or contrast in which it is uniquely present.

[53]

**Design-contrast identifiability is necessary but not sufficient: the three Gompertz trajectories are collinear, so joint estimation is credible only under specific design and sample-size conditions.**

- The components share an onset shape and overlap in calendar time, so their parameters can be near-collinear even when the identifying contrasts are present.
- Joint recovery requires a phase in which each component varies while the others are held fixed, and an off-drug window long enough for  $TV$  to separate from a decaying  $BR$ .
- The recoverable region of the design-and-sample-size space is mapped by the pre-registered identifiability diagnostics, not asserted.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Identifiability in the design-contrast sense above is necessary but not sufficient for stable estimation at finite sample sizes. The three Gom-

pertz trajectories share an onset shape and overlap in calendar time, so their parameters can be collinear even when the contrasts that identify them in principle are present. Two conditions make the joint estimation credible: the design must supply at least one phase in which each component varies while the others are held fixed (the blinded-placebo contrast for  $BR$ , the open-label-versus-blinded contrast for  $PB$ , and an off-drug window for  $TV$ ), and that off-drug window must be long enough relative to the carryover half-life for  $TV$  to separate from a still-decaying  $BR$ . Where these conditions are weak the components are jointly identified only up to a near-collinear direction, and the symptom of that is the rank-deficiency the pilot of section 7 encountered when the full phase-augmented formula was fitted at  $N = 35$ . We therefore do not assert that the three components are always recoverable. The pre-registered identifiability diagnostics of section 7 (convergence rates of the component-matched fit, variance-inflation factors on the component-specific Gompertz parameters, and the sensitivity of the  $BR$  estimate to the assumed  $\eta(\text{phase})$  multipliers) are the means by which the recoverable region of the design-and-sample-size space is to be mapped, and the framework's participant-level claims should be read as conditional on falling inside that region.

## 5 Analysis-side methodology

[54]

**Four constraints make a component-matched analysis impractical at trial-relevant sample sizes; a phase-augmented LME captures the important part of the gap at minimal cost.**

- The question of whether to mirror the DGP in the analysis model is reasonable; the section addresses it directly.
- The four constraints are identifiability, degrees of freedom, convergence fragility, and marginal estimand robustness.
- A phase-by-treatment indicator is the recommended modest extension that captures BR-versus-PB sensitivity without the full parametric cost.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

In this section we shall take up the practical question of what analysis model to use. A reasonable reaction to the formal model in section 2 is the following: if we are simulating data with three components plus noise, why is the analysis model written as  $Sx \sim \text{bm} + \text{t} + \text{Dbc} + \text{bm}:\text{Dbc}$  with a single random intercept and AR(1) residual correlation? Should we not include three parametric Gompertz components in the analysis, mirroring the DGP, so that the analyst sees the

1469 same structure that generated the data? The question is reasonable.  
1470 Four constraints make a component-matched analysis impractical for  
1471 most aggregated N-of-1 trial sizes, and a modest extension (a phase-  
1472 by-treatment indicator) captures the practically important part of the  
1473 gap at minimal cost.

## 1474 5.1 Four reasons not to match the DGP

1475 [55]

1476 **The first constraint is identifiability: forcing a three-**  
1477 **component model on a design with insufficient contrasts**  
1478 **produces unidentified parameters, not component estimates.**

- 1479 • A pure open-label trial cannot distinguish  $BR$  from  $PB$ ; both  
1480 rise from zero together on treatment initiation.
- 1481 • Fitting three components to a one-contrast design produces im-  
1482 plausibly tight standard errors that misrepresent inferential pre-  
1483 cision.
- 1484 • The analyst must audit whether the design supplies the necessary  
1485 contrasts before specifying a component-matched model.

1486 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
1487 *tempor incididunt ut labore et dolore magna aliqua.*

1488 The first reason is identifiability. The trial design must supply the  
1489 contrasts that identify each component, and many designs supply only  
1490 part of it. The decomposition is identified by phase contrasts (open-  
1491 label versus blinded for  $PB$ , on-drug versus blinded-placebo for  $BR$ ,  
1492 off-drug timepoints for  $TV$ ). A pure open-label trial has none of these  
1493 contrasts; in such a trial  $BR$  and  $PB$  both rise from zero on treat-  
1494 ment initiation and are statistically indistinguishable from each other,  
1495 regardless of how the analyst writes the model. Forcing an analysis to  
1496 fit three components when the design supplies contrasts for only one  
1497 or two of them does not recover the missing components; it produces  
1498 unidentified or weakly identified parameter estimates with implausibly  
1499 tight standard errors that misrepresent the analyst's actual inferential  
1500 precision.

1501 [56]

1502 **The second constraint is degrees of freedom: a component-**  
1503 **matched analysis adds approximately a dozen non-linear**  
1504 **parameters, inflating moderation estimand variance at**  
1505  **$N \in [30, 150]$ .**

- 1506 • Gompertz  $BR$  (3),  $PB$  (3 + expectancy multiplier), and  $TV$  (3)  
1507 each require participant-specific random effects on the maxima.

- 1508       • At trial-relevant  $N$ , the variance of  $\hat{\beta}_{bm:D}$  is dominated by partic-  
1509       participant and timepoint counts.  
1510       • Additional parametric structure typically inflates moderation es-  
1511       timand variance more than it reduces bias.

1512       *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
1513       *tempor incididunt ut labore et dolore magna aliqua.*

1514       The second reason is degrees of freedom. Each component term  
1515       costs degrees of freedom that come out of the moderation estimand.  
1516       The biomarker-moderation question (the core scientific aim of the  
1517       predictive-biomarker validation programme) is asked through the  
1518       bm:Dbc term. A component-matched analysis adds a Gompertz  
1519       BR (three parameters), a Gompertz PB (three parameters plus a  
1520       phase-specific expectancy multiplier), and a Gompertz TV (three  
1521       parameters), each with participant-specific random effects on the  
1522       maxima. That is on the order of a dozen new parameters, half of  
1523       them on a non-linear scale. At trial sample sizes of  $N \in [30, 150]$ , the  
1524       variance of  $\hat{\beta}_{bm:D}$  is dominated by participant and timepoint counts,  
1525       and adding parametric structure to absorb residual confounding  
1526       usually inflates the moderation estimand's variance by more than it  
1527       reduces bias.

1528       [57]

1529       **The third constraint is convergence fragility: component-**  
1530       **matched nonlinear mixed models converge below 90% at**  
1531       **trial-relevant  $N$ , contaminating aggregated analyses.**

- 1532       • Simultaneous Gompertz BR, PB, and TV fitting requires  
1533       nlme::nlme, Bayesian hierarchical, or structural-equation  
1534       approximations.  
1535       • Each is more sensitive to starting values, prone to local optima,  
1536       and likely to fail on individual trial datasets than the LME ana-  
1537       logue.  
1538       • Non-converging cells at below-90% convergence rates contami-  
1539       nate power and bias estimates in simulation studies.

1540       *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
1541       *tempor incididunt ut labore et dolore magna aliqua.*

1542       The third reason is convergence fragility. Component-matched anal-  
1543       yses are non-linear in the parameters and their convergence is prob-  
1544       lematic in practice. Fitting Gompertz BR, PB, and TV simulta-  
1545       neously requires a non-linear mixed model (nlme::nlme with self-  
1546       starting Gompertz forms), a Bayesian hierarchical fit, or a structural-  
1547       equation approximation. Each is more sensitive to starting values,  
1548       more prone to local optima, and more likely to fail to converge on a  
1549       single trial dataset than the linear-mixed analogue. At trial-relevant

$N$ , the convergence rate of the component-matched fit is typically below 90%, and the non-converging cells contaminate any aggregated analysis.

[58]

**The fourth constraint is estimand robustness: the  $\text{bm:D}_{bc}$  marginal interaction is the right estimand for the biomarker-validation question regardless of the BR-PB split.**

- Regulators want to know whether the biomarker predicts treatment response under trial conditions, not a mechanistically decomposed prediction.
- The marginal  $\text{bm:D}_{bc}$  interaction captures this question whether the moderation is biological or expectation-mediated.
- Component decomposition adds mechanistic precision but is not required for the core biomarker-validation decision.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The fourth reason is that the moderation estimand is sufficiently robust to the BR-versus-PB split for the most common scientific question. For predictive-biomarker validation, the regulator wants to know whether the biomarker predicts treatment response under the trial conditions, not a clean pharmacological-versus-placebo decomposition of that prediction. The simple  $\text{bm:D}_{bc}$  interaction is the right marginal estimand for the validation question whether the moderation is biological or expectation-mediated.

[59]

**The inferential target is stated as an estimand: the population biomarker-by-treatment interaction with explicit intercurrent-event handling, while the pharmacological-component target is more ambitious and inherits the heterogeneity-identifiability caveat.**

- The  $\text{bm:D}_{bc}$  estimand is defined by its population, treatment contrast, endpoint, summary, and intercurrent-event strategy in the ICH E9(R1) sense.
- Dropout is handled by a treatment-policy strategy and residual carryover by the modelled exponential decay.
- Recovering a participant-level pharmacological interaction inherits Senn's caveat that true heterogeneity is separable from within-person variation only under repeated phase contrasts.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

It is worth stating the inferential target of the analysis in the vocabulary of the estimands framework<sup>53</sup>, because the biomarker-validation question is easy to pose ambiguously. The primary estimand is the population-level biomarker-by-treatment interaction, the  $bm:D_{bc}$  coefficient: its population is the trial-eligible patient population, its treatment contrast is on-drug versus off-drug exposure summarised through the continuous indicator  $D_{bc}$ , its endpoint is the symptom trajectory, its population-level summary is the interaction coefficient, and its intercurrent events are handled by a treatment-policy strategy for dropout and by the modelled exponential decay for residual carryover. The pharmacological-component target  $\beta_{bm}^{BR}$  is a more ambitious estimand: it is the biomarker-by- $BR$  interaction that would obtain with the belief-driven channel held at its off-drug level, identified here through the phase contrasts of section 4 rather than by assumption. We flag explicitly that recovering a participant-level interaction of this kind inherits the difficulty Senn has emphasised for individual treatment effects<sup>49,54</sup>: a genuine participant-by-component signal is separable from participant-by-period and within-person random variation only when the design supplies repeated phase contrasts, and the framework should be read as estimating these quantities under that condition rather than as dissolving the well-known problem of identifying individual effects from finite within-person data.

## 5.2 A small extension that usually pays off: phase indicators

[60]

**The phase-augmented LME uses `Dbc:phase` and `bm:Dbc:phase` to test whether drug and biomarker-by-drug effects attenuate under blinding.**

- `phase` absorbs systematic differences in  $PB$  expectancy across trial phases.
- `Dbc:phase` detects whether the drug effect shrinks under blinding, a signature of  $PB$ -mediation.
- `bm:Dbc:phase` detects whether the biomarker-by-treatment interaction shrinks under blinding, a signature that it is partly biomarker-by- $PB$ .

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The smallest useful step toward component-aware analysis, without paying the costs above, is to add a phase indicator and its interaction with the drug indicator. Schematically,

```
Sx ~ bm + t + Dbc + phase + Dbc:phase
      + bm:Dbc + bm:Dbc:phase
      + (1 | ptID), correlation = corCAR1(...)
```

Here `phase` is a categorical covariate (open-label, blinded, post-crossover) that absorbs the systematic differences in *PB* expectancy across phases. `Dbc:phase` tests whether the apparent drug effect shrinks under blinding, a signature that the open-label drug effect was partly *PB*-mediated. `bm:Dbc:phase` tests whether the moderation itself shrinks under blinding, a signature that the biomarker-by-treatment interaction was partly biomarker-by-*PB* rather than biomarker-by-*BR*.

[61]

**The phase-augmented extension adds at most four parameters yet captures the practically relevant belief-attenuation diagnostic without requiring the Gompertz parametric form.**

- Three to four parameters is well within budget at any realistic trial sample size.
- The extension does not separately estimate *BR* and *PB* components but identifies the change in apparent drug effect when belief is hedged.
- For most analyses this is the practically relevant question and is answerable without committing to the Gompertz form.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

This extension adds three to four parameters at most, well within budget at any realistic  $N$ . It does not separately estimate *BR* and *PB* as components, but it does identify the change in apparent drug effect that occurs when belief is hedged. For most analyses, this is the practically relevant question, and it can be answered without committing to the Gompertz parametric form. Where both the lumped and the phase-augmented analyses are reported, the additional `Dbc:phase` and `bm:Dbc:phase` terms introduce a small family of belief-attenuation tests, and the attribution inference they support should be treated as a pre-specified secondary analysis with its multiplicity acknowledged rather than as a second bite at the primary interaction test.

### 5.3 When to fit the full decomposition

[62]

**The full decomposition merits the effort only when  $N$  is in the hundreds, the design is fully phase-balanced, and the scientific question requires the BR-PB mechanistic split.**

- Feasibility conditions: several-hundred  $N$ , dense timepoints in every phase, mechanistic BR-PB split required, and informative Bayesian priors available.

- In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or `brms` can return participant-specific component estimates.
- Outside that regime the phase-augmented LME is the better tool.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

A component-matched analysis merits the effort when sample size is in the several hundreds (a phase-3 trial fielded as aggregated N-of-1, or a meta-analysis pooling multiple smaller trials); the trial design provides timepoints in every phase combination, with enough density per cell to constrain the Gompertz curves; the scientific question requires the BR-PB split (a pharmacology paper claiming a specific mechanism, where ruling out a placebo contribution is the central point); and a Bayesian hierarchical fit with informative priors on the Gompertz parameters is feasible. In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or a `brms` hierarchical formulation can return participant-specific *BR*, *PB*, and *TV* component estimates that mirror the DGP. Outside that regime, the linear-mixed analysis with phase indicators is the better tool.

[63]

**The DGP-analysis asymmetry is deliberate: simulation uses the rich DGP to characterise how a simpler analysis behaves; the analysis need not mirror the DGP to produce useful inference.**

- Simulation controls the DGP and can specify any structure; inference must pay the identifiability cost of every parameter.
- The DGP is a characterisation tool; the analysis is a finite-data inference tool; they have different optimality criteria.
- DGP-matching is not required and is often counterproductive at trial-relevant sample sizes.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The asymmetry between DGP and analysis is not a flaw of the framework; it is a deliberate consequence of the asymmetry between simulation (where the analyst controls the DGP and can specify any structure, however elaborate) and inference (where the analyst must fit a model to a finite dataset and pay the identifiability cost of every additional parameter). Simulation studies use the rich DGP to characterise how a simpler analysis behaves under realistic data generation; they do not require the analysis to match the DGP in order to produce useful inference.

[64]

The exponential-decay carryover indicator is a deliberately parsimonious special case of richer carryover models, and the simulation should benchmark it against a distributed-lag alternative.

- The continuous indicator  $D_{bc}$  decays as  $\exp(-\lambda t_{sd})$ , a one-parameter form tied to a pharmacokinetic half-life.
- Distributed-lag models, the average-period-treatment-effect estimand, and G-estimation relax the single-rate assumption at additional parametric cost.
- The exponential restriction is an assumption, not a finding, and the pre-registered simulation should include a distributed-lag arm to quantify its cost.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

A word on the carryover parameterisation is in order, since it is the one place where the analysis model makes a strong functional-form commitment. The continuous drug indicator  $D_{bc}$  decays as  $\exp(-\lambda t_{sd})$  off drug, a one-parameter exponential governed by the carryover half-life. This is deliberately parsimonious and is best understood as a constrained special case of the more flexible carryover models now available for repeated-measures and N-of-1 data. Liao et al.<sup>31</sup> model carryover through a Bayesian distributed-lag structure with autocorrelated errors, relaxing the single-rate exponential assumption at the cost of additional parameters; Daza<sup>55</sup> formalises an average-period treatment-effect estimand built to accommodate autocorrelation, trend, and slow carryover jointly; and G{”a}rtner et al.<sup>56</sup> benchmark linear models against G-estimation and Bayesian-network alternatives specifically under carryover. We adopt the exponential form because it ties directly to a pharmacokinetic half-life and conserves degrees of freedom for the moderation estimand at trial-relevant sample sizes, but the choice is an assumption rather than a finding. The pre-registered simulation programme of section 7 should accordingly include a distributed-lag analysis arm, so that the cost of the exponential-decay restriction is quantified against the richer alternative rather than asserted.

## 6 Worked example: information loss without decomposition

### 6.1 What the lumped analysis estimates

[65]

The lumped treatment-effect estimate has a probability limit equal to the pharmacological component plus belief and

**natural-history contributions, so its bias does not vanish as  $N$  grows.**

- Writing the reduction as  $BR + PB + TV - \varepsilon$  and fitting one drug indicator gives  $\text{plim } \hat{\beta}_D = m_{BR} + b_{PB} + b_{TV}$ .
- $b_{PB}$  and  $b_{TV}$  are the belief and natural-history terms that load onto the drug indicator under the chosen contrast; both are generally non-negative for an improving outcome.
- Because the displacement is in the probability limit, a larger one-component trial converges on the same biased value: the failure is identification, not precision.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Before the numerical illustration it is worth establishing algebraically what the one-component analysis actually estimates, because the qualitative failure modes of section 1 are consequences of an identity rather than empirical tendencies that require simulation to demonstrate. Write the within-participant symptom reduction as  $R_{it} = BL_i - Y_{it} = BR_{it} + PB_{it} + TV_{it} - \varepsilon_{it}$ . The one-component analysis fits a single drug indicator  $D_{it}$ ,

$$R_{it} = \alpha + \beta_D D_{it} + g(t) + u_i + e_{it},$$

and reports  $\hat{\beta}_D$  as ‘the treatment effect’. Taking the on-drug minus off-drug contrast that  $\beta_D$  targets, and using that the natural-history component is by construction independent of treatment status so that its on/off difference cancels in a within-participant comparison, the probability limit of the estimator is

$$\text{plim } \hat{\beta}_D = m_{BR} + b_{PB} + b_{TV},$$

where  $b_{PB} = E[PB \mid \text{on}] - E[PB \mid \text{off}]$  is the belief differential between the contrasted phases and  $b_{TV}$  collects any natural-history contribution that fails to cancel under the contrast actually used;  $b_{TV}$  is zero for a clean on/off within-participant contrast and strictly positive for a baseline-anchored pre-post analysis, in which case  $b_{PB} = E[PB]$  and  $b_{TV} = E[TV]$ . To fix which case is in play below: the one-component analysis used in the worked example and the simulations of this paper is the design-contrast (drug-on/off) form, so  $b_{TV} \approx 0$  there and the operative displacement is the belief term  $b_{PB}$ ; the baseline-anchored pre-post form, which additionally carries  $b_{TV} = E[TV]$ , is the worse case retained only for comparison. Both terms are generally non-negative for an improving outcome, so the lumped estimate over-states the pharmacological component  $m_{BR}$ . The displacement

is a property of the probability limit and is unchanged as  $N \rightarrow \infty$ : a larger one-component trial converges on the same biased value. This is the precise sense in which the placebo-attribution failure of section 1 is a problem of identification, not of precision.

[66]

**The biomarker slope from a lumped analysis decomposes additively into component-specific slopes, so a biomarker correlated with  $PB$  or  $TV$  registers as a predictor even with no pharmacological association.**

- By linearity of covariance,  $\beta_{bm}^{\text{lumped}} = \beta_{bm}^{BR} + w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV}$ .
- A non-zero  $\beta_{bm}^{PB}$  or  $\beta_{bm}^{TV}$  produces a non-zero lumped slope even when the pharmacological slope  $\beta_{bm}^{BR}$  is exactly zero.
- The result is an algebraic identity under the additive decomposition, not a sample-size-dependent tendency.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The same algebra settles the biomarker-validation failure. The precision-medicine analysis regresses a participant-level response summary on a baseline biomarker  $B_i$ . If that summary is the lumped per-participant effect, which estimates  $m_{BR,i} + w_{PB}m_{PB,i} + w_{TV}m_{TV,i}$  for design-determined loadings  $w_{PB}, w_{TV} \geq 0$ , then by the linearity of covariance the fitted biomarker slope has probability limit

$$\beta_{bm}^{\text{lumped}} = \frac{\text{Cov}(m_{BR} + w_{PB}m_{PB} + w_{TV}m_{TV}, B)}{\text{Var}(B)} = \beta_{bm}^{BR} + w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV},$$

where  $\beta_{bm}^C = \text{Cov}(m_C, B)/\text{Var}(B)$  is the component-specific slope. A biomarker correlated with placebo responsiveness or with the natural-history trajectory, so that  $\beta_{bm}^{PB}$  or  $\beta_{bm}^{TV}$  is non-zero, therefore registers a non-zero lumped slope even when its pharmacological association  $\beta_{bm}^{BR}$  is exactly zero. This is the biomarker-validation failure of section 1, and it too is an identity under the additive decomposition rather than an empirical tendency: no sample size separates the lumped slope from a genuine pharmacological one, because the two coincide in the estimand whenever  $w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV} \neq 0$ .

We make two qualifications explicit, because they bound how much weight the framework can carry. First, both displays are immediate corollaries of the additive decomposition and the linearity of the projection that defines the estimator; they are not a derived result in any deeper sense, and their value is in naming the displacement terms, not in the algebra. Second, they assume the components combine additively. Under the subadditive interaction documented in section 3.3,

where the joint effect of  $BR$  and  $PB$  falls below their sum, a first-order correction proportional to  $\text{Cov}(m_{BR} \cdot m_{PB}, B)$  enters the biomarker slope, and the identities above should be read as the leading behaviour in the additive case rather than as exact under interaction. The empirical magnitude of that correction term is itself an open question that the framework's  $\gamma$ -interaction simulation (section 3.3) is designed to address.

[67]

**The bias displaces the estimand rather than inflating variance, so a significance test of the lumped coefficient cannot detect it; the decomposition is what locates it.**

- The lumped coefficient is displaced from  $m_{BR}$  by  $b_{PB} + b_{TV}$  but remains a well-estimated quantity with a valid Wald test of its own nullity.
- Detecting the bias requires comparing  $\hat{\beta}_D$  to  $m_{BR}$ , a comparison only the decomposition or an isolating phase contrast supplies.
- This is why a rejection-rate summary is the wrong instrument, and accounts for the flat rejection rates across the placebo-strength axis in the section 7.1 pilot.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Two consequences for how the framework should be evaluated follow. First, the quantity that carries the bias is the location of the estimand,  $\text{plim } \hat{\beta}_D$ , not its sampling variability: the lumped coefficient remains a well-estimated number with a valid Wald test of its own nullity, and that test rejects at the rate dictated by the displaced value. A significance test of the lumped coefficient therefore cannot, in principle, detect the bias, because the bias does not move the coefficient toward zero. Detecting it requires comparing  $\hat{\beta}_D$  against  $m_{BR}$ , which is exactly the comparison the decomposition, or a phase contrast that isolates  $m_{BR}$ , makes available and the lumped analysis does not. Second, and as a corollary for the simulation evidence, a rejection-rate or power summary is the wrong instrument for this bias: the right target is the coefficient itself and its offset from  $m_{BR}$ . This is why the pilot of section 7.1 found the biomarker-by-treatment rejection rate essentially flat across the placebo-strength axis even though that axis is precisely what drives the predicted bias; the prediction lives in the mean of  $\hat{\beta}_{bm:D_{bc}}$ , and a production run that means to test it must read the bias off the coefficient, not off the test.

## 6.2 A numerical illustration

[68]

The worked example uses two participants with identical lumped responses but opposite BR-PB profiles to show that the one-component analysis cannot distinguish them.

- Participant A has high  $PB$  ( $m_{PB} = 6$ ) and moderate  $BR$  ( $m_{BR} = 4$ ); Participant B has low  $PB$  ( $m_{PB} = 1$ ) and high  $BR$  ( $m_{BR} = 8$ ).
- Both participants reach approximately 10–11 points of total improvement by week 8, making them indistinguishable to a one-component analysis.
- The blinded-discontinuation phase creates the diagnostic contrast that exposes the underlying difference.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

To illustrate the inferential cost of failing to decompose, we shall work through a simple numerical example. Consider two participants, A and B, enrolled in a sixteen-week N-of-1 trial of prazosin for PTSD. Both are otherwise similar: same age, same baseline severity, same trial design. They differ in two latent traits that the trial does not measure directly. Participant A is a high placebo responder and a moderate pharmacological responder, with true component values  $m_{BR}^A = 4$ ,  $m_{PB}^A = 6$ ,  $m_{TV}^A = 1$ . Participant B is a low placebo responder and a strong pharmacological responder, with  $m_{BR}^B = 8$ ,  $m_{PB}^B = 1$ ,  $m_{TV}^B = 1$ .

[69]

**Open-label totals are clinically indistinguishable; the one-component analysis estimates a population drug effect of approximately 10 points with no differentiation between A and B.**

- Both participants show approximately 10–11 points of improvement and present as clinical successes.
- A t-test or simple LME on open-label change cannot detect the heterogeneity in  $BR$  versus  $PB$  composition.
- The population drug-effect estimate conflates both pharmacological and placebo contributions.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

In the open-label phase (weeks 1 to 8), both  $BR$  and  $PB$  run at full strength and  $TV$  accumulates. By week 8 both participants have reached close to their saturation values:

|               | <i>BR</i> | <i>PB</i> ( $\eta = 1$ ) | <i>TV</i> | Total improvement |
|---------------|-----------|--------------------------|-----------|-------------------|
| Participant A | 4.0       | 6.0                      | 1.0       | <b>11.0</b>       |
| Participant B | 8.0       | 1.0                      | 1.0       | <b>10.0</b>       |

A clinician examining only the totals would conclude that the two participants are responding similarly: both improved by about ten points, both look like clinical successes. A simple t-test on the open-label change would estimate a population drug effect of about ten points and would not distinguish the two participants.

[70]

**The blinded-discontinuation phase reveals the BR-PB heterogeneity: Participant A retains 4 points while B collapses to 1.5, exposing the diagnostic contrast.**

- Participant A’s high *PB* at  $\eta = 0.5$  (3 points) sustains improvement even without active drug.
- Participant B’s low *PB* (0.5 points) and absent *BR* produce near-complete return to baseline.
- The 4.0 versus 1.5 contrast is exactly what the decomposition uses to separate *BR* from *PB*.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

In the blinded discontinuation phase (weeks 9 to 12), both participants are silently switched to placebo at the start of week 9. The expectancy multiplier drops from 1.0 to 0.5. For clarity of exposition we take the carryover half-life to be short relative to the four-week blinded window, so that *BR* has effectively cleared by the end of week 12; the more realistic case of a slowly decaying *BR*, modelled by the carryover machinery of section 3.1, attenuates but does not eliminate the contrast below. By the end of week 12, the components are:

|               | <i>BR</i> | <i>PB</i> ( $\eta = 0.5$ ) | <i>TV</i> | Total improvement |
|---------------|-----------|----------------------------|-----------|-------------------|
| Participant A | 0.0       | 3.0                        | 1.0       | <b>4.0</b>        |
| Participant B | 0.0       | 0.5                        | 1.0       | <b>1.5</b>        |

The two participants now look quite different. Participant A still shows a sizeable improvement (four points) under blinded placebo; participant B has nearly returned to baseline (one and a half points). This is the diagnostic contrast that the decomposition exploits. For simplicity the table holds *TV* at its week-8 value; in reality *TV* continues to accumulate over weeks 9 to 12, which shifts both totals by a

1940 common amount and leaves the *BR*-versus-*PB* contrast on which the  
1941 example turns unchanged.

1942 [71]

1943 **The one-component model estimates drug at 7.75 points with**  
1944 **no principled decomposition, obscuring that A's true *BR* is**  
1945 **4 and B's is 8.**

- 1946 • The 7.75 point estimate is a conflated average of true *BR* values  
1947 (4.0 and 8.0) plus the placebo wash-out.
- 1948 • The model cannot determine whether A and B differ, nor estimate  
1949 the pharmacological drug effect separately.
- 1950 • Placebo and natural-history effects are not identifiable from this  
1951 contrast alone.

1952 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
1953 *tempor incididunt ut labore et dolore magna aliqua.*

1954 A one-component fixed-effects regression of total symptom change on  
1955 phase-by-allocation indicators sees:

- 1956 • Open-label means: 11.0 (A), 10.0 (B).
- 1957 • Blinded-placebo means: 4.0 (A), 1.5 (B).
- 1958 • Estimated drug effect:  $11.0 - 4.0 = 7.0$  (A),  $10.0 - 1.5 = 8.5$  (B).  
1959 Population mean: 7.75.
- 1960 • Estimated placebo effect: not identifiable from this contrast.
- 1961 • Estimated natural history: not identifiable from this contrast.

1962 The one-component model has produced a single number conflated  
1963 with the placebo response that washes out under blinding. It cannot  
1964 say whether the drug effect is the same for participants A and B; it  
1965 estimates 'drug' at 7.75 points on average, which is an average of the  
1966 true 4.0 (A) and 8.0 (B) mixed with the placebo wash-out, with no  
1967 principled way to decompose further.

1968 [72]

1969 **The full decomposition recovers all six component estimates**  
1970 **by exploiting phase differential expectancy and the blinded-**  
1971 **placebo contrast.**

- 1972 • The LME returns  $\hat{m}_{BR}^A = 4.0$ ,  $\hat{m}_{BR}^B = 8.0$ ,  $\hat{m}_{PB}^A = 6.0$ ,  $\hat{m}_{PB}^B =$   
1973  $1.0$  at the population point estimate.
- 1974 • Participant-level estimates have meaningful uncertainty at trial-  
1975 relevant  $N$ ; the simulation study characterises this.
- 1976 • The phase differential ( $\eta = 1$  vs.  $\eta = 0.5$ ) is the identification  
1977 mechanism for the *PB* parameter.

1978 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
1979 *tempor incididunt ut labore et dolore magna aliqua.*

A linear-mixed-effects analysis with the full BR-PB-TV decomposition, fitted to the same data, can recover the individual components by exploiting the differential expectancy between phases and the on-drug-versus-blinded-placebo contrast. After fitting, the model returns  $\hat{m}_{BR}^A = 4.0$ ,  $\hat{m}_{BR}^B = 8.0$ ,  $\hat{m}_{PB}^A = 6.0$ ,  $\hat{m}_{PB}^B = 1.0$ ,  $\hat{m}_{TV}^A = \hat{m}_{TV}^B = 1.0$  (at the population point estimate; participant-level estimates have meaningful uncertainty at  $N$  in the trial-relevant range, which the simulation study below characterises).

[73]

**The one-component model produces one biased average; the three-component model produces six clinically actionable quantities, making biomarker validation possible.**

- The lumped estimate of 7.75 points is a biased average of  $BR_A = 4$  and  $BR_B = 8$  confounded with placebo wash-out.
- The decomposition supplies individual  $BR$ ,  $PB$ , and  $TV$  estimates enabling per-participant clinical decisions.
- Predictive-biomarker validation requires the decomposition because biomarker-by- $BR$  is the target estimand.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The information loss from lumping the components is not abstract. We may compare the inferences directly:

| Question                                  | One-component answer       | Three-component answer                                                                                                         |
|-------------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Does the drug work for A?                 | ~7.75 points (lumped).     | $BR_A = 4.0$ , distinct from her larger placebo response.                                                                      |
| Does the drug work for B?                 | ~7.75 points (lumped).     | $BR_B = 8.0$ , twice the magnitude of A.                                                                                       |
| Are A and B different responders?         | No discernible difference. | B has 2x A's $BR$ but 1/6 of A's $PB$ .                                                                                        |
| Is a biomarker predicting BR useful here? | Cannot answer.             | Yes if the biomarker correlates with $\hat{m}_{BR}$ .                                                                          |
| Should A be continued long-term?          | Cannot say.                | Probably yes ( $BR = 4$ is real and pharmacological), but the larger placebo component will not persist if expectations decay. |

| Question                        | One-component<br>answer | Three-component<br>answer                            |
|---------------------------------|-------------------------|------------------------------------------------------|
| Population mean<br>drug effect? | 7.75 points (biased).   | $6.0 = (4.0 + 8.0)/2$<br>(correct mean of<br>$BR$ ). |

The one-component model has produced one biased average where the three-component model has produced six clinically actionable quantities. Predictive-biomarker validation, which is a central scientific aim of the trial design family considered here, becomes impossible without the decomposition.

[74]

**A population-level example shows that cohort differences in open-label totals driven entirely by  $PB$  produce apparent non-replicability that the decomposition resolves.**

- Cohort 1 (naive,  $m_{PB} = 7$ ) reports 13-point improvements; Cohort 2 (sceptical,  $m_{PB} = 2$ ) reports 8 points; true  $m_{BR} = 5$  in both.
- A one-component analysis would falsely conclude the drug works less well in Cohort 2 and trigger searches for confounders.
- The decomposition returns  $\hat{m}_{BR} = 5$  in both cohorts, attributing the difference correctly to  $PB$  heterogeneity.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

A second example at the population level illustrates the same problem. Consider two cohorts of N-of-1 participants enrolled in methodologically identical trials of the same drug. Cohort 1 (early trial, naive patients) has mean  $m_{BR} = 5$ , mean  $m_{PB} = 7$ , mean  $m_{TV} = 1$ . Cohort 2 (later trial, patients who have had multiple prior failed treatments and are sceptical) has mean  $m_{BR} = 5$ , mean  $m_{PB} = 2$ , mean  $m_{TV} = 1$ . By construction the drug works equally well in both cohorts. But the open-label totals differ substantially: cohort 1 reports thirteen-point improvements on average, cohort 2 reports eight-point improvements. A one-component analysis comparing the two cohorts would conclude that the drug works less well in cohort 2 and might trigger a search for confounders or sub-population effects. None of this is real; the difference is entirely in  $PB$ , specifically in the population baseline level of placebo responsiveness. The three-component decomposition, applied to either cohort, returns  $\hat{m}_{BR} = 5$  in both. Unfortunately, this is the kind of apparent non-replicability that pharmacology has spent the last decade struggling to navigate; the decomposition is one of the formal tools that makes it tractable.

## 7 Simulation study

[75]

**The simulation study follows ADEMP and quantifies bias and identifiability properties of three analysis strategies across a prazosin-PTSD-calibrated parameter grid.**

- The Morris (2019) ADEMP framework structures the study: Aims, DGP, Estimands, Methods, Performance measures.
- The pre-registration is at `analysis/scripts/component-decomposition/00-ademp-pre-registration`.
- Three analysis strategies (one-component, phase-augmented, full decomposition) are compared across the parameter grid.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The simulation study reported and pre-registered in this section is structured under the<sup>57</sup> ADEMP framework, specifying Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. The pre-registered plan is at `analysis/scripts/component-decomposition/00-ademp-pre-registration.md`. A Monte Carlo simulation calibrated to the prazosin/PTSD reference parameters quantifies the bias and identifiability properties of the three analysis strategies described in section 5 across a parameter grid.

### 7.1 Pilot validation of analysis machinery (100 replicates, 6 cells)

[76]

**The pilot is a structural check of the analysis pipeline, not a substantive empirical result.**

- The realised Study A run (section 7.3) is the empirical core of the manuscript; this pilot precedes it.
- The pilot confirms only that data generation, model fitting, and summary output run without error.
- No bias or power claims are derived from the 100-replicate pilot.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The production simulation programme is the empirical core of this manuscript; its first realised run, Study A, is reported in section 7.3, while the fuller pre-registered grid (section 7.2) remains partly outstanding. The pilot reported here precedes both and confirms only that the analysis machinery runs end-to-end.

[77]

The pilot executed a 6-cell, 100-replicate run with perfect convergence, confirming the pipeline is operational.

- The grid crossed three  $pb\_strength$  levels against two analysis strategies.
- Convergence was 1.0 across all six cells under 8-core `mclapply`.
- Output is checkpointed at `analysis/data/quick-sim/`; no inferential claims are drawn.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

A 100-replicate-per-cell pilot was run at the reference Hybrid OL+BDC design,  $N = 35$ , on the 6-cell slice  $pb\_strength \in \{0, 6.5, 13\} \times analysis \in \{\text{one-component, phase-augmented}\}$ , using 8-core `parallel::mclapply`. The convergence rate was 1.0 across all six cells. Per-replicate output is checkpointed at `analysis/data/quick-sim/06-decomp-replicates.rds` and the Morris-format cell-level summary at `analysis/data/quick-sim/06-decomp-summary.txt`. The exploratory summaries are diagnostic only. The one-component `bm:Dbc` test showed power 0.95, 0.93, and 0.83 across  $pb\_strength \in \{0, 6.5, 13\}$ , while the phase-augmented analysis showed 0.65, 0.65, and 0.61; the mean estimated  $\hat{\beta}_{bm:D_{bc}}$  was essentially flat within each analysis (about  $-2.25$  for the one-component and  $-2.21$  for the phase-augmented fit). At 100 replicates the Monte Carlo standard error on a power estimate near 0.7 is about 0.046, so these figures cannot by themselves separate the two analyses; their interpretation, including why the phase-augmented analysis loses its bias-detection contrast at this  $N$ , is resolved by the production run of section 7.3. No confirmatory bias or power claim is drawn from the pilot, and all substantive claims are deferred to the production programme below.

[78]

**Two pilot findings revise the production specification: rank-deficiency forces a formula change, and bias must be read from coefficient means, not rejection rates.**

- At  $N = 35$ , rank-deficiency dropped the  $D_{bc}:\text{phase}$  term, eliminating the key biomarker-by-treatment-by-phase contrast.
- The predicted bias manifests in the mean of  $\hat{\beta}_{bm:D_{bc}}$ , not in the Wald-test rejection rate.
- The production programme addresses both by using larger  $N$ , a fuller design, and coefficient-mean reporting.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Two design-level observations from the pilot inform the production

specification. First, at  $N = 35$  the rank-deficiency of the full phase-augmented formula required dropping the  $D_{bc}$ :phase interaction term, removing the precise contrast that would have isolated the biomarker-by-treatment-by-phase signal the framework is designed to detect. Second, the predicted bias of the one-component analysis arises in the regression coefficient  $\hat{\beta}_{bm:D_{bc}}$ , not in the indicator-level rejection rate, so power on the biomarker-by-treatment Wald test is not the right summary statistic to detect the framework's predicted bias. The production programme below addresses both observations through larger  $N$ , a richer trial design that admits the full phase-augmented formula, and reporting of the coefficient mean against the implied true value rather than the rejection rate alone.

## 7.2 Production-simulation specification

[79]

**The production programme follows the ADEMP framework with four deliverables; the realised Study A run in section 7.3 is the first of these.**

- The deliverables are interaction-estimate bias, full-decomposition identifiability, type I error and power, and participant-level signal recovery.
- Sample sizes  $N \in \{35, 70, 100, 150\}$  cross a *pb\_strength* grid; type I cells target MCSE 0.003 at 5,000 replicates.
- Driver scripts, checkpointed outputs, and the ADEMP pre-registration reside under `analysis/scripts/component-decomposition/`.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The production programme follows the<sup>57</sup> ADEMP framework and targets four deliverables: the bias of  $\hat{\beta}_{bm:D_{bc}}$  for the one-component, phase-augmented, and full-decomposition analyses across  $N \in \{35, 70, 100, 150\}$  and a *pb\_strength* grid; identifiability metrics for the full-decomposition fit (convergence, variance-inflation factors, and sensitivity of the *BR* estimate to the  $\eta$ (phase) multipliers); type I error and power for the interaction test under an  $\eta$ -mismatch, *TV*-magnitude, and dropout sensitivity grid, with null cells at  $n_{\text{reps}} = 5,000$  for MCSE 0.003; and the participant-level interaction signal recovered as a function of  $(m_{BR}, m_{PB}, m_{TV})$ , which the lumped-total regression does not preserve. The `tidyverse` implementation collection drives the grid; driver scripts, checkpointed cell-level summaries, per-replicate output, and the committed ADEMP pre-registration all reside under `analysis/scripts/component-decomposition/`. The realised Study A run reported next used 1,000 replicates per alternative cell.

### 7.3 Study A results: bias and power at 1,000 replicates per alternative cell

[80]

**At 1,000 replicates per alternative cell the one-component estimate of  $\hat{\beta}_{bm:D_{bc}}$  is essentially unbiased across the entire  $(m_{PB}, m_{TV})$  grid, contradicting the framework's prediction of a placebo-strength-driven bias.**

- The true biomarker-by- $BR$  effect is  $-2.25$ ; mean absolute bias is  $0.016$  at  $N = 35$ ,  $0.016$  at  $N = 70$ ,  $0.006$  at  $N = 100$ , and  $0.006$  at  $N = 150$ , in every case within one Monte Carlo SE of zero.
- Bias shows no monotone trend across  $m_{PB} \in \{0, 1, 3, 6, 10\}$  or  $m_{TV} \in \{-1, 0, 1, 2\}$ ; it behaves as sampling noise around zero.
- This is the predicted consequence of the identity in section 6.1: the DGP couples the biomarker to  $BR$  only, so  $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$  and the lumped slope is unbiased by construction.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The production Study A run (1,000 replicates per alternative cell, 5,000 per null cell, 80 alternative cells crossing  $m_{PB} \in \{0, 1, 3, 6, 10\}$ ,  $m_{TV} \in \{-1, 0, 1, 2\}$ , and  $N \in \{35, 70, 100, 150\}$ , plus four null cells) returns a result that does not support the framework's central bias prediction. The data-generating process couples the baseline biomarker to the  $BR$  component only, through the on-drug biomarker- $BR$  covariance channel, so the true biomarker-by-treatment effect is the biomarker-by- $BR$  slope,  $\beta_{bm:D_{bc}} = -c_{bm} \sigma_{BR} / \sigma_{bm} = -2.25$ , and the biomarker is by construction uncorrelated with  $PB$  and  $TV$ . Under the one-component analysis the mean estimated  $\hat{\beta}_{bm:D_{bc}}$  tracks this value across the whole grid: the mean absolute bias is  $0.016$  at  $N = 35$ ,  $0.016$  at  $N = 70$ ,  $0.006$  at  $N = 100$ , and  $0.006$  at  $N = 150$ , in every case within one Monte Carlo standard error (respectively about  $0.028$ ,  $0.019$ ,  $0.009$ , and  $0.007$ ) of zero, with no monotone trend across  $m_{PB}$  or  $m_{TV}$ . The predicted placebo-attribution bias of the biomarker-by-treatment estimate does not appear.

This is not a failure of the simulation; it is the behaviour the identity of section 6.1 requires. That identity writes the lumped biomarker slope as  $\beta_{bm}^{BR} + w_{PB} \beta_{bm}^{PB} + w_{TV} \beta_{bm}^{TV}$ . With the biomarker coupled to  $BR$  alone,  $\beta_{bm}^{PB} = \beta_{bm}^{TV} = 0$ , so the lumped slope equals the pharmacological slope no matter how large  $m_{PB}$  or  $m_{TV}$  grows. Study A varies the magnitudes of the unmodelled components, which the identity shows to be the innocuous axis; the axis that produces biomarker-validation bias is the biomarker's correlation with  $PB$  or  $TV$ , which this DGP holds at zero. The clean null is therefore a confirmation of the identity

and a correction to the looser claim of section 1 that large  $PB$  or  $TV$  by itself biases biomarker validation. It does not; biomarker-component coupling does.

[81]

**The phase-augmented analysis provides no bias reduction and incurs a substantial power penalty, so it is dominated under this uncontaminated DGP.**

- Mean power at  $N = 35$  is 0.35 (phase-augmented) versus 0.59 (one-component); at  $N = 70$ , 0.62 versus 0.90; both reach approximately 1.0 by  $N = 100$ .
- Phase-augmented mean absolute bias is no smaller than one-component at any  $N$  (0.028 versus 0.016 at  $N = 35$ ).
- Convergence is approximately 100% for both analyses at every  $N$  (minimum 0.998), so the rank-deficiency that degraded the  $N = 35$  pilot does not recur here.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The comparison between analyses is equally clear. The phase-augmented analysis is not less biased than the one-component analysis at any sample size (its mean absolute bias is 0.028 versus 0.016 at  $N = 35$  and is comparable elsewhere), and it is markedly less powerful for the biomarker-by-treatment test: mean power across the grid is 0.35 versus 0.59 at  $N = 35$  and 0.62 versus 0.90 at  $N = 70$ , with both analyses reaching essentially full power by  $N = 100$ . Because the DGP contains no biomarker- $PB$  contamination for the phase contrast to remove, the extra phase-by-treatment terms add estimation cost without buying any bias reduction, and the analysis is dominated. Convergence was approximately complete for both analyses at every sample size (minimum convergence rate 0.998), so the rank-deficiency that forced dropping the `Dbc:phase` term in the  $N = 35$  pilot, which we had advanced as the explanation for the pilot's inverted power ranking, is not operative in the production run: the inverted ranking persists with the full formula fitted and converged. The pilot's power inversion was therefore a genuine feature of this DGP, not an artifact of rank-deficiency.

Calibration is adequate for both analyses. Nominal-95% confidence-interval coverage of  $\beta_{bm:D_{bc}}$  ranges from 0.96 to 0.99 for the one-component analysis (slightly conservative) and from 0.94 to 0.98 for the phase-augmented analysis. Type I error at the four null cells is controlled, ranging from 0.031 to 0.056 for the one-component analysis and 0.038 to 0.049 for the phase-augmented analysis; the one-component estimate at  $N = 100$  (0.056, MCSE 0.003) sits 1.9 standard errors above nominal but is within the simulation's Monte Carlo

uncertainty.

[82]

**Study A confirms the identity but cannot exhibit the biomarker-validation failure; a contaminated-biomarker pilot reported in the next subsection provides the decisive evidence.**

- As pre-registered, Study A varies component magnitude with the biomarker fixed to  $BR$ , so it probes an axis the identity predicts to be null.
- The decisive axis is  $c_{bm,PB}$ , the biomarker- $PB$  correlation; this is varied in the contaminated-biomarker pilot below.
- On current evidence the phase-augmented analysis is not recommended for the uncontaminated-biomarker regime; the pilot below evaluates it under contamination.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Two conclusions follow for the Study A programme. First, the result sharpens rather than supports the section 1 framing: the biomarker-validation failure is real as an identity, but it is driven by the biomarker's coupling to  $PB$  or  $TV$ , not by the magnitude of those components, and Study A as pre-registered varies the latter while holding the former at zero. The decisive axis is the biomarker- $PB$  correlation  $c_{bm,PB}$ , which the contaminated-biomarker pilot in the following subsection varies directly. Second, on the present evidence the phase-augmented analysis carries a real power cost without a compensating bias reduction under an uncontaminated biomarker, so it should not be recommended as a default for this design; the pilot below evaluates whether contamination changes this verdict.

#### 7.4 Contaminated-biomarker pilot (100 replicates per cell)

[83]

**A pilot varying the biomarker- $PB$  correlation  $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$  confirms that bias in the one-component estimator scales with coupling strength, not component magnitude, and that the phase-augmented analysis provides no protection.**

- At  $N = 150$ , mean  $\hat{\beta}_{bm:D_{bc}}$  shifts from  $-2.25$  (true) to  $-1.74$  at  $c_{bm,PB} = 0.45$  (bias  $+0.51$ , 30 Monte Carlo SEs from zero), confirming the identity's prediction.
- The bias is driven by  $c_{bm,PB} \times \sigma_{PB}$ , not by the population-mean  $m_{PB}$ : cells with  $m_{PB} = 0$  and  $m_{PB} = 6$  show identical bias

at every  $c_{bm,PB}$  level, because the individual-level  $PB$  variance ( $\sigma_{PB} = 2$ ) provides the contamination channel even when the mean is zero.

- The phase-augmented analysis shows the same bias as the one-component analysis at every contamination level, offering no protection against biomarker- $PB$  coupling.
- This pilot uses 100 replicates per cell; a production run at 1,000 replicates per cell is required to confirm the pattern and assess type I error under contamination.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

A supplementary pilot varying the biomarker- $PB$  correlation  $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$  at two  $PB$  magnitude levels ( $m_{PB} \in \{0, 6\}$ ) and two sample sizes ( $N \in \{70, 150\}$ ), using 100 replicates per cell, confirms the identity-based prediction and reveals an unexpected structural result.

At  $N = 150$  the one-component bias scales monotonically with  $c_{bm,PB}$ : mean  $\hat{\beta}_{bm:D_{bc}}$  moves from  $-2.31$  ( $c_{bm,PB} = 0$ , bias  $-0.06$ ) to  $-2.37$  ( $c_{bm,PB} = 0.15$ , bias  $-0.12$ ) to  $-2.08$  ( $c_{bm,PB} = 0.30$ , bias  $+0.17$ ) to  $-1.74$  ( $c_{bm,PB} = 0.45$ , bias  $+0.51$ ). The last value sits 30 Monte Carlo standard errors from the true  $-2.25$ , establishing the effect unambiguously. The direction of bias is positive (toward zero): high- $c_{bm,PB}$  participants experience stronger  $PB$ , which in the open-label phase adds to the apparent drug effect, but this elevation is partially absorbed by the  $bm$  main effect, leaving the  $bm : D_{bc}$  interaction coefficient attenuated relative to the pure pharmacological value.

The unexpected result is that  $m_{PB}$  does not modulate the bias: cells with  $m_{PB} = 0$  and  $m_{PB} = 6$  produce identical bias at every contamination level (differences  $\leq 0.006$  at  $N = 150$ , well within Monte Carlo error). The driver is  $c_{bm,PB} \times \sigma_{PB}$ , not  $c_{bm,PB} \times m_{PB}$ . Even when the population-mean  $PB$  amplitude is zero, individual participants have nonzero  $PB$  draws (standard deviation  $\sigma_{PB} = 2$ ), and coupling the biomarker to that variance is sufficient to contaminate the estimator. A true no- $PB$  control requires  $\sigma_{PB} = 0$ , not  $m_{PB} = 0$ . This refines the Study A design: varying  $m_{PB}$  at  $c_{bm,PB} = 0$  correctly produced zero bias, but because the relevant null was  $c_{bm,PB} = 0$ , not  $m_{PB} = 0$ .

The phase-augmented analysis provides no protection: its bias at every  $(c_{bm,PB}, m_{PB}, N)$  cell is indistinguishable from the one-component bias. Adding phase-by-treatment interactions to the model does not decompose the contamination because both analyses see the biomarker- $PB$  covariance through the same open-label versus blinded-discontinuation contrast; the contrast changes the weight on  $PB$  (via expectancy) but does not isolate the  $PB$  component's

contribution to the biomarker slope. Separating that contribution requires the full nonlinear component decomposition (Study B), which is pending implementation.

The  $N = 70$  results are uninformative at 100 replicates (Monte Carlo standard error  $\approx 0.06$ , comparable to or larger than the bias at moderate  $c_{bm,PB}$ ). The production run will use 1,000 replicates per cell to establish the  $N = 70$  pattern and 5,000 null replicates to assess type I error under contamination. Until the production run is complete, the quantitative bias estimates from this pilot should be treated as directionally informative but imprecise.

[84]

**The contaminated-biomarker pilot reframes the paper’s practical guidance: the risk of biomarker-validation failure arises from biomarker- $PB$  correlation in the study population, not from the presence of strong placebo response per se.**

- A trial in a population where the biomarker is uncorrelated with placebo responsiveness can use the one-component estimator without bias, regardless of how large the placebo response is.
- A trial where the biomarker covaries with optimism, suggestibility, or treatment-seeking history is at material risk of contamination; bias of  $+0.51$  (23% of the true effect) at  $c_{bm,PB} = 0.45$  illustrates the practical scale.
- Neither the one-component nor the phase-augmented analysis removes this bias; full component decomposition is required.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The pilot results reframe the paper’s practical guidance. The risk of biomarker-validation failure in an aggregated N-of-1 study arises not from the presence of a strong placebo response in the population but from the biomarker’s correlation with individual-level placebo responsiveness. In a population where the baseline biomarker is uncorrelated with each participant’s placebo-belief response amplitude, the one-component estimator recovers the pharmacological interaction without bias, and the full decomposition machinery is not required. In a population where the biomarker covaries with psychological characteristics that predict placebo response – optimism, prior treatment experience, suggestibility, expectancy at enrolment – the one-component analysis conflates pharmacological and placebo-belief prediction, and the scale of that conflation grows with the coupling strength. At  $c_{bm,PB} = 0.45$ , a value that equals the pharmacological coupling used throughout this paper, the one-component bias is approximately 0.51 points on the interaction scale, representing 23% of the true pharmacological effect. This is a practically meaningful distortion for any precision-medicine

stratification decision.

The production run of the contaminated-biomarker study, at 1,000 alternative-cell replicates and 5,000 null-cell replicates, is required to confirm these estimates, establish the  $N = 70$  pattern, and characterise type I error under contamination. Results are pending.

## 8 Application to predictive-biomarker validation

[85]

**The precision-medicine question – whether a biomarker predicts pharmacological response – is answered by the lumped total only when the biomarker is uncorrelated with  $PB$  and  $TV$ ; otherwise it requires decomposing the response.**

- The question is whether the biomarker predicts the  $BR$  component, not the lumped total.
- The PATH statement formalises this as the target estimand for predictive biomarker validation.
- The BR-PB-TV decomposition is the mechanism that separates the pharmacological signal when biomarker-component coupling cannot be ruled out.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The motivating clinical question for the trial designs in this project is whether a baseline biomarker predicts treatment response, and specifically whether the biomarker predicts the pharmacological component of treatment response. This is the precision-medicine question, formalised in the PATH statement<sup>14</sup>. Whether it depends on the BR-PB-TV decomposition is itself conditional: by the identity of section 6.1, the lumped biomarker slope equals the pharmacological slope when the biomarker is uncorrelated with  $PB$  and  $TV$ , and departs from it only by the biomarker's covariance with those components. The decomposition is therefore the mechanism the question requires precisely in the regime where that covariance is non-negligible.

[86]

**A lumped-total biomarker regression conflates pharmacological and placebo-belief prediction, producing a stratification biomarker that mis-identifies responders.**

- $\hat{\beta}_{bm}$  from the lumped regression contains both pharmacological and placebo-belief prediction in unknown proportions.
- If  $PB$  covaries with the biomarker distribution, flagged 'high responders' include high- $PB$  responders who will not benefit phar-

macologically.

- Prospective application of such a biomarker would systematically mis-stratify patients for prescription decisions.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

A direct regression of total response on baseline biomarker

```
total_response = beta_0 + beta_bm * biomarker + error
```

estimates the biomarker's correlation with whatever the lumped total contains. If  $PB$  varies systematically across the biomarker distribution, because the biomarker is correlated with optimism, suggestibility, or treatment-seeking history, then  $\hat{\beta}_{bm}$  contains a placebo-prediction component inseparable from the pharmacological-prediction component. The resulting biomarker, applied prospectively, would mis-stratify patients: those flagged as 'high responders' would include both true high- $BR$  responders and high- $PB$  responders, and the latter would not benefit from prescription decisions made on a pharmacological basis.

[87]

**A valid pharmacological predictor requires a non-zero  $\hat{\beta}_{bm}^{BR}$ ; significant  $\hat{\beta}_{bm}^{PB}$  or  $\hat{\beta}_{bm}^{TV}$  indicate a psychological or prognostic biomarker, not a pharmacological one.**

- The sole criterion for pharmacological prediction is a meaningfully non-zero  $BR$ -component regression coefficient.
- A biomarker correlating only with  $PB$  selects placebo responders, not drug responders.
- A biomarker correlating only with  $TV$  is prognostic but not predictive in the pharmacological sense.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The right approach is to regress each component on the biomarker:

$$\begin{aligned} BR &= \beta_0^{BR} + \beta_{bm}^{BR} \cdot \text{biomarker} + \varepsilon^{BR}, \\ PB &= \beta_0^{PB} + \beta_{bm}^{PB} \cdot \text{biomarker} + \varepsilon^{PB}, \\ TV &= \beta_0^{TV} + \beta_{bm}^{TV} \cdot \text{biomarker} + \varepsilon^{TV}. \end{aligned}$$

A biomarker is a useful pharmacological predictor if and only if  $\hat{\beta}_{bm}^{BR}$  is meaningfully non-zero. Significant  $\hat{\beta}_{bm}^{PB}$  or  $\hat{\beta}_{bm}^{TV}$  are diagnostic findings about the biomarker, indicating that it is partly psychological or correlates with natural-history characteristics, but they are not the

validation of pharmacological prediction. A biomarker that correlates only with  $PB$  would be valuable for selecting placebo responders in run-in designs but worthless for predicting pharmacological efficacy; a biomarker that correlates only with  $TV$  would be a prognostic marker, not a predictive one.

[88]

**When biomarker-component coupling cannot be ruled out, the decomposition is the mechanism by which the precision-medicine question is posed; when it can, the lumped analysis already answers it.**

- Where the biomarker may correlate with  $PB$  or  $TV$ , separating  $BR$  from them is what prevents validation from conflating pharmacological and non-pharmacological prediction.
- The phase-augmented `bm:Dbc:phase` interaction is a small- $N$  proxy, but Study A shows it buys nothing under an uncontaminated biomarker, so it is warranted only when contamination is plausible.
- The full decomposition recovers component-specific regressions explicitly when sample size permits.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The decomposition is therefore the mechanism by which the precision-medicine question is posed whenever the candidate biomarker may correlate with  $PB$  or  $TV$ . Where the biomarker can be assumed orthogonal to the non-pharmacological components, the identity of section 6.1 shows the lumped analysis already targets the biomarker-by- $BR$  estimand, and the simulation of section 7.3 confirms it recovers that estimand without bias. The phase-augmented linear-mixed analysis introduced in section 5 retains a useful subset of the decomposition machinery at small  $N$  (the `bm:Dbc:phase` interaction tests whether the biomarker-by-treatment effect attenuates under blinding, the practical proxy for biomarker-by- $PB$  contamination), but Study A shows this proxy earns its degrees of freedom only when such contamination is present: under an uncontaminated biomarker it reduced no bias and lost power. The full decomposition recovers the component-specific regressions explicitly when  $N$  permits and when the scientific question warrants the cost.

## 9 Discussion

[89]

**The three-component framework reframes treatment re-**

sponse as a structured sum of causally distinct contributions, and the present work maps when acting on that decomposition actually changes inference.

- The framework specifies which trial-design contrasts identify which components.
- Acting on the decomposition is valuable conditionally: for attribution estimands, and for biomarkers that may correlate with  $PB$  or  $TV$ , but not for a biomarker-by-treatment interaction with a biomarker orthogonal to them.
- The target estimand for biomarker validation is the biomarker-by- $BR$  association; whether a lumped analysis already recovers it depends on the biomarker's coupling to  $PB$  and  $TV$ .

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

The three-component decomposition is the formal expression of the claim that ‘treatment response’ in an aggregated N-of-1 trial is not a single quantity but a sum of three causally distinct contributions, each with its own pharmacological, psychological, or epidemiological interpretation. What the present work establishes, analytically in section 6.1 and empirically in section 7.3, is that the inferential value of acting on this decomposition is conditional rather than universal. The identity shows that the one-component estimator is displaced from its pharmacological target only by the components’ loading onto the contrast used, and, for a biomarker slope, only by the biomarker’s covariance with  $PB$  and  $TV$ . Where those terms are zero, the design-contrast one-component analysis is already unbiased, and the production simulation confirms it: with a biomarker coupled to  $BR$  alone, the one-component biomarker-by-treatment estimate is unbiased across the whole  $PB-TV$  grid and the phase-augmented analysis is dominated. The framework’s contribution is therefore best read as a map of when decomposition changes inference, not as a blanket prescription to decompose. It makes the trial-design conditions explicit, identifies the attribution and contaminated-biomarker estimands for which decomposition is needed, and clarifies the target estimand for biomarker validation (the biomarker-by- $BR$  association, not the lumped total).

[90]

**Four open questions bound the framework’s current scope: the simulation results are pending, the  $\eta$  multipliers require sensitivity analysis, non-monotone natural history is not yet handled, and participant-level recovery inherits the identifiability caveats of individual treatment effects.**

- Quantitative bias-versus-power trade-offs are conjectures until the section 7 simulation is run.

- The  $\eta$ (phase) expectancy-modulation parameters are modelling assumptions requiring explicit sensitivity analysis.
- Cyclical or biphasic  $TV$  trajectories exceed the Gompertz form and require flexible or covariate-augmented specifications.
- Participant-level component recovery rests on separating participant-by-component signal from participant-by-period and participant-by-phase variation, the identifiability caveat Senn raises for individual treatment effects.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Four open questions remain and warrant acknowledgement. First, the simulation study described in section 7 is forthcoming; the present manuscript reports the framework rather than its empirical evaluation, and the quantitative bias-versus-power trade-offs across analysis strategies are conjectures until that work is run. Second, the  $\eta$ (phase) multipliers used to parameterise the expectancy modulation of  $PB$  are themselves modelling assumptions, and a sensitivity analysis varying  $\eta$  across plausible values is mandatory before any clinical conclusion is drawn from the framework’s outputs. Third, the framework as specified treats the three components as monotone Gompertz trajectories; cyclical or biphasic  $TV$  patterns require either flexible  $TV$  specifications or explicit covariates beyond the Gompertz form, and the boundary between ‘flexible-Gompertz’ and ‘covariate-augmented’ specifications is not yet sharply drawn. Fourth, the participant-level component estimates the framework targets inherit the identifiability caveat that<sup>54</sup> and<sup>49</sup> raise for individual treatment effects more generally: within-participant component recovery rests on separating a true participant-by-component signal from participant-by-period and participant-by-phase variation, and this separation is credible only when the design supplies repeated phase contrasts and the cross-component covariance structure is not pathological. The framework should be read as estimating these quantities under those assumptions, not as dissolving the well-known difficulty of estimating individual effects from finite within-person data.

[91]

**Two companion documents extend the manuscript: a pedagogical narrative and a Gompertz-family sensitivity evaluation.**

- The pedagogy document at `docs/24-component-decomposition-pedagogy.md` provides worked examples for first-time readers.
- The Gompertz evaluation at `analysis/report/07-gompertz-evaluation/` quantifies trajectory-specification sensitivity.
- Together they address the intuitive and parametric dimensions of

2570 the framework that the present manuscript treats briefly.

2571 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
2572 *tempor incididunt ut labore et dolore magna aliqua.*

2573 The companion pedagogical document at docs/24-component-decomposition-pedagogy.md  
2574 develops the framework with worked examples and an extended in-  
2575 tuition narrative for readers approaching the material for the first  
2576 time. The companion methodological evaluation of the Gompertz  
2577 family at analysis/report/07-gompertz-evaluation/ quantifies  
2578 the sensitivity of the framework's inference to the specific parametric  
2579 form chosen for the component trajectories.

2580 [92]

2581 **The components exist in any longitudinal treatment re-**  
2582 **sponse, but participant-level identifiability requires a**  
2583 **within-subject belief-by-exposure contrast that not every**  
2584 **placebo-controlled design supplies.**

- 2585 • The three-component structure is a property of the data-  
2586 generating process of any repeated-measures treatment study;  
2587 its identifiability is design-specific.
- 2588 • Parallel-group placebo control identifies the population-mean  $PB$   
2589 but not a participant-level  $PB$ , and supplies no within-subject  
2590 belief contrast; only designs with such a contrast (blinded or  
2591 open-hidden discontinuation, randomised withdrawal, crossover,  
2592 hybrid N-of-1) identify the participant-level decomposition the  
2593 biomarker question needs.
- 2594 • The broad claim is therefore scoped to designs supplying a within-  
2595 subject belief-by-exposure contrast, not to placebo-controlled lon-  
2596 gitudinal trials in general.

2597 *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod*  
2598 *tempor incididunt ut labore et dolore magna aliqua.*

2599 A final point concerns the scope of the decomposition, where it is  
2600 essential to distinguish the existence of the components from their  
2601 identifiability. The claim that an observed longitudinal treatment  
2602 response is the sum of a pharmacological, a belief-driven, and a  
2603 natural-history component is a statement about the data-generating  
2604 process of any repeated-measures treatment study, and in that  
2605 sense the decomposition is not specific to N-of-1 trials. Identifi-  
2606 ability, by contrast, is design-specific, and the distinction between  
2607 population-level and participant-level recovery is decisive for the  
2608 biomarker application. A standard parallel-group placebo-controlled  
2609 trial supplies a between-arm placebo contrast that identifies the  
2610 population-mean belief response but not a participant-level  $PB$ , and  
2611 it contains no within-subject contrast in which belief varies indepen-

dently of pharmacology; the participant-level biomarker-by- $BR$  slope that the precision-medicine question targets is therefore not identified from such a design, no matter how large it is. The designs that do supply the required within-subject belief contrast are the blinded or open-hidden discontinuation and randomised-withdrawal designs<sup>51,52</sup>, the crossover with on/off periods, and the hybrid N-of-1 design developed here; the pharmacometric disease-progression-plus-placebo models from which the framework descends<sup>1-3</sup> operate at the group level and recover population-mean components only. We therefore advocate the decomposition as a default lens for longitudinal designs that supply a within-subject belief-by-exposure contrast, with the aggregated N-of-1 case the most demanding instance because it must recover the components from a single participant's trajectory. We do not claim participant-level identifiability from parallel-group placebo control alone, and where only a between-arm placebo contrast is available the framework's claims are restricted to the population mean.

## 10 Conclusions

[93]

**Treatment response in aggregated N-of-1 trials is the sum of three causally distinct components, and the bias from conflating them is governed by the estimand and the biomarker's coupling to the components, not by their raw magnitude.**

- The three components are  $BR$  (pharmacological),  $PB$  (placebo-belief), and  $TV$  (natural history).
- Each has its own biological mechanism, identifiability conditions, and clinical interpretation.
- The lumped pre-post treatment effect is biased by the  $PB$  and  $TV$  contributions; the biomarker-by-treatment slope is biased only by the biomarker's covariance with  $PB$  and  $TV$ , as Study A confirms.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

In conclusion, three points are to be emphasised. First, treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components: pharmacological action ( $BR$ ), placebo-belief response ( $PB$ ), and time-variant natural history ( $TV$ ). Each component has its own biological mechanism, its own identifiability conditions, and its own clinical interpretation. Whether conflating them biases a given analysis is, however, governed by the omitted-variable-bias identity of section 6.1, not by the raw magnitude of  $PB$  and  $TV$ : the

lumped pre-post treatment effect is displaced by the  $PB$  and  $TV$  contributions, whereas the biomarker-by-treatment slope is displaced only by the biomarker's covariance with those components. Study A bears this out: varying  $PB$  and  $TV$  magnitude with a biomarker coupled to  $BR$  alone left the one-component estimate unbiased across the entire grid.

[94]

**What matters is the design contrast, not the parametric model: the trial must supply the drug-on/off and belief contrasts, but adding parametric component terms is not free and is not always worth it.**

- Unique component identification requires phase contrasts built into the trial design; the one-component analysis already exploits the drug-on/off contrast.
- Under an uncontaminated biomarker, the phase-augmented analysis adds no bias reduction and loses power, so it is not recommended as a default.
- The phase-augmented and full component-matched analyses are warranted for attribution estimands and for biomarkers that may correlate with  $PB$  or  $TV$ , where the extra structure does work.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Second, the operative requirement is the design contrast, not the parametric analysis model. The trial design must include the drug-on/off and belief contrasts that identify the components, and the one-component analysis already exploits the drug-on/off contrast through its  $D_{bc}$  term. Adding parametric component structure on top is not free: under an uncontaminated biomarker the phase-augmented analysis reduced no bias and lost substantial power, so it should not be adopted as a default. Its value, and that of the full component-matched fit, is confined to the estimands where the extra structure does work, namely the attribution of response to pharmacology versus belief and the analysis of biomarkers that may themselves correlate with  $PB$  or  $TV$ .

[95]

**For predictive-biomarker validation the decomposition matters precisely when the candidate biomarker may correlate with  $PB$  or  $TV$ ; this is the case the decisive simulation still has to probe.**

- A biomarker that predicts the lumped total is a pharmacological predictor only if it is uncorrelated with  $PB$  and  $TV$ ; otherwise the lumped slope is contaminated.

- Regulatory and prescribing decisions therefore require separating  $BR$  from  $PB$  and  $TV$  whenever biomarker-component coupling cannot be ruled out.
- Study A established the uncontaminated baseline; the contaminated-biomarker cell is the planned empirical test of the failure mode.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Third, for predictive-biomarker validation the decomposition earns its place precisely when the candidate biomarker may correlate with  $PB$  or  $TV$ . A biomarker that predicts the lumped total is a pharmacological predictor only when it is uncorrelated with the non-pharmacological components; where that cannot be assumed, the lumped slope mixes pharmacological and non-pharmacological prediction, and the regulatory and prescribing decisions that motivate the trial require the separation the decomposition supplies. The empirical demonstration of this failure mode, a data-generating process that couples the biomarker to  $PB$ , is the decisive cell that Study A did not include and that remains the natural next step; the present run establishes the uncontaminated baseline against which it should be compared.

## 11 Conflict of interest

The authors declare no conflicts of interest.

## 12 Funding

This work received no specific funding.

## 13 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

## 14 Reproducibility

[96]

**The manuscript is rendered inside a Docker container with locked package versions; the full software environment is pinned by Dockerfile and renv.lock.**

- R 4.4.x in the zzcollab Docker container provides the execution environment.
- Simulation packages are `nlme`, `parallel`, `data.table`, `furrr`, and `purrr`.
- The manuscript build uses `rmarkdown` and `bookdown`.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the original-extended simulation collection), `furrr` and `purrr` (the tidyverse simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

[97]

**Reproducibility is ensured by a master seed of 20260507 with per-replicate seeds derived by index addition.**

- `set.seed(20260507)` is called at the top of each replicate loop.
- `parallel::mc.set.seed` propagates seeds inside `mclapply`.
- Per-replicate seeds are `master_seed + rep_idx`.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

[98]

**All simulation outputs, driver scripts, pre-registration, and manuscript source files are versioned at documented paths within the repository.**

- Per-replicate outputs and Morris summaries are at `analysis/data/quick-sim/`.
- Driver scripts and the ADEMP pre-registration are at `analysis/scripts/component-decomposition/`.
- Manuscript source, bibliography, and CSL file are at `analysis/report/06-component-decomposition/`.

*Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.*

Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/06-decomp-replicates`  
 Morris ADEMP cell-level summaries at the companion `06-decomp-summary.txt`.  
 Driver scripts are under `analysis/scripts/component-decomposition/`.

The pre-registered ADEMP plan is at `analysis/scripts/component-decomposition/00-ademp-pre-reg`  
 The manuscript source, paper-specific bibliography (`references.bib`),  
 and SIM CSL file are at `analysis/report/06-component-decomposition/`.

## 15 References

1. Holford N. Clinical pharmacology = disease progression + drug action. *British Journal of Clinical Pharmacology*. 2015;79(1):18-27. doi:10.1111/bcp.12170
2. Gomeni R, Lavergne A, Merlo-Pich E. Modelling placebo response in depression trials using a longitudinal model with informative dropout. *European Journal of Pharmaceutical Sciences*. 2009;36(1):4-10. doi:10.1016/j.ejps.2008.10.025
3. Chen C, Jönsson S, Yang S, Plan EL, Karlsson MO. Detecting placebo and drug effects on Parkinson's disease symptoms by longitudinal item-score models. *CPT: Pharmacometrics & Systems Pharmacology*. 2021;10(4):309-317. doi:10.1002/psp4.12601
4. Muthén B, Brown HC. Estimating drug effects in the presence of placebo response: Causal inference using growth mixture modeling. *Statistics in Medicine*. 2009;28(27):3363-3385. doi:10.1002/sim.3721
5. Kaptechuk TJ, Kelley JM, Conboy LA, et al. Components of placebo effect: Randomised controlled trial in patients with irritable bowel syndrome. *BMJ*. 2008;336(7651):999-1003. doi:10.1136/bmj.39524.439618.25
6. Kirsch I, Deacon BJ, Huedo-Medina TB, Scoboria A, Moore TJ, Johnson BT. Initial severity and antidepressant benefits: A meta-analysis of data submitted to the Food and Drug Administration. *PLoS Medicine*. 2008;5(2):e45. doi:10.1371/journal.pmed.0050045
7. Locher C, Kossowsky J, Gaab J, Kirsch I, et al. Moderation of antidepressant and placebo outcomes by baseline severity in late-life depression: A systematic review and meta-analysis. *Journal of Affective Disorders*. 2015;181:50-60. doi:10.1016/j.jad.2015.03.062
8. Sugarman MA, Loree AM, Baltes BB, Grekin ER, Kirsch I. The efficacy of paroxetine and placebo in treating anxiety and depression. *PLoS ONE*. 2014;9(8):e106337. doi:10.1371/journal.pone.0106337
9. Hall KT, Lembo AJ, Kirsch I, et al. Catechol-O-methyltransferase val158met polymorphism predicts placebo effect in irritable bowel syndrome. *PLoS ONE*. 2012;7(10):e48135. doi:10.1371/journal.pone.0048135

- 2784 10. Wang RS, Lembo AJ, Kaptchuk TJ, Hall KT, et al. Genomic effects associated with response to placebo treatment in a randomized trial of irritable bowel syndrome. *Frontiers in Pain Research*. 2022;2:775386. doi:10.3389/fpain.2021.775386
- 2785 11. Hróbjartsson A, Gøtzsche PC. Is the placebo powerless? An analysis of clinical trials comparing placebo with no treatment. *New England Journal of Medicine*. 2001;344(21):1594-1602. doi:10.1056/NEJM200105243442106
- 2786 12. Hróbjartsson A, Gøtzsche PC. Is the placebo powerless? Update of a systematic review with 52 new randomized trials. *Journal of Internal Medicine*. 2004;256(2):91-100. doi:10.1111/j.1365-2796.2004.01355.x
- 2787 13. Hróbjartsson A, Gøtzsche PC. Placebo interventions for all clinical conditions. *Cochrane Database of Systematic Reviews*. 2010;(1):CD003974. doi:10.1002/14651858.CD003974.pub3
- 2788 14. Kent DM, Paulus JK, Klavere D van, et al. The Predictive Approaches to Treatment effect Heterogeneity (PATH) Statement. *Annals of Internal Medicine*. 2020;172(1):35-45. doi:10.7326/M18-3667
- 2789 15. Kent DM, Klavere D van, Paulus JK, et al. The PATH statement: Explanation and elaboration. *Annals of Internal Medicine*. 2020;172(1):W1-W25. doi:10.7326/M18-3668
- 2790 16. Kent DM, Steyerberg E, Klavere D van. Personalized evidence based medicine: Predictive approaches to heterogeneous treatment effects. *BMJ*. 2018;363:k4245. doi:10.1136/bmj.k4245
- 2791 17. Rekkas A, Paulus JK, Raman G, et al. Predictive approaches to heterogeneous treatment effects: A scoping review. *BMC Medical Research Methodology*. 2020;20(1):264. doi:10.1186/s12874-020-01145-1
- 2792 18. Guyatt G, Sackett D, Taylor DW, Chong J, Roberts R, Pugsley S. Determining optimal therapy: Randomized trials in individual patients. *New England Journal of Medicine*. 1986;314(14):889-892.
- 2793 19. Zucker DR, Schmid CH, McIntosh MW, D'Agostino RB, Selker HP, Lau J. Combining single patient (N-of-1) trials to estimate population treatment effects and to evaluate individual patient responses to treatment. *Journal of Clinical Epidemiology*. 1997;50(4):401-410. doi:10.1016/S0895-4356(96)00429-5

- 2794 20. Zucker DR, Ruthazer R, Schmid CH. Individual (N-of-1) trials can be combined to give population comparative treatment effect estimates: Methodologic considerations. *Journal of Clinical Epidemiology*. 2010;63(12):1312-1323. doi:10.1016/j.jclinepi.2010.04.020
- 2795 21. Lillie EO, Patay B, Diamant J, Issell B, Topol EJ, Schork NJ. The n-of-1 clinical trial: The ultimate strategy for individualizing medicine? *Personalized Medicine*. 2011;8(2):161-173. doi:10.2217/pme.11.7
- 2796 22. Duan N, Kravitz RL, Schmid CH. Single-patient (n-of-1) trials: A pragmatic clinical decision methodology for patient-centered comparative effectiveness research. *Journal of Clinical Epidemiology*. 2013;66(8 Suppl):S21-S28. doi:10.1016/j.jclinepi.2013.04.006
- 2797 23. Gabler NB, Duan N, Vohra S, Kravitz RL. N-of-1 trials in the medical literature: A systematic review. *Medical Care*. 2011;49(8):761-768. doi:10.1097/MLR.0b013e318215d90d
- 2798 24. Schork NJ, Goetz LH. Single-subject studies in translational nutrition research. *Annual Review of Nutrition*. 2017;37:395-422. doi:10.1146/annurev-nutr-071816-064717
- 2799 25. Schork NJ. Randomized clinical trials and personalized medicine: A commentary on Deaton and Cartwright. *Social Science and Medicine*. 2018;210:71-73. doi:10.1016/j.socscimed.2018.04.033
- 2800 26. Schork NJ, Beaulieu-Jones B, Liang WS, Smalley S, Goetz LH. Exploring human biology with N-of-1 clinical trials. *Cambridge Prisms: Precision Medicine*. 2023;1:e12. doi:10.1017/pcm.2022.15
- 2801 27. Porcino A, Vohra S. N-of-1 trials, their reporting guidelines, and the advancement of open science principles. *Harvard Data Science Review*. 2022;4(SI3). doi:10.1162/99608f92.a65a257a
- 2802 28. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 statement. *Journal of Clinical Epidemiology*. 2015;76:9-17. doi:10.1016/j.jclinepi.2015.05.004
- 2803 29. Shamseer L, Sampson M, Bukutu C, Schmid CH, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015: Explanation and elaboration. *Journal of Clinical Epidemiology*. 2015;76:18-46. doi:10.1016/j.jclinepi.2015.05.018

- 2804 30. Li J, Gao W, Punja S, et al. Reporting quality of N-of-1 trials published between 1985 and 2013: A systematic review. *Journal of Clinical Epidemiology*. 2016;76:57-64. doi:10.1016/j.jclinepi.2015.11.016
- 2805 31. Liao Z, Qian M, Kronish IM, Cheung YK. Analysis of N-of-1 trials using Bayesian distributed lag model with autocorrelated errors. *Statistics in Medicine*. 2023;42(13):2044-2060. doi:10.1002/sim.9676
- 2806 32. Blackston JW, Chapple AG, McGree JM, McDonald S, Nikles J. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. *Healthcare*. 2019;7(4):137. doi:10.3390/healthcare7040137
- 2807 33. Carrozzo AE, Zimmermann G, Bathke AC, et al. Two-arm crossover randomized controlled trial versus meta-analysis of N-of-1 studies: Comparison of statistical efficiency. *Biometrical Journal*. 2025;67(2):e70045. doi:10.1002/bimj.70045
- 2808 34. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and practical considerations. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013
- 2809 35. Kaptchuk TJ, Friedlander E, Kelley JM, et al. Placebos without deception: A randomized controlled trial in irritable bowel syndrome. *PLoS ONE*. 2010;5(12):e15591. doi:10.1371/journal.pone.0015591
- 2810 36. Lembo A, Kelley JM, Nee J, et al. Open-label placebo vs double-blind placebo for irritable bowel syndrome: A randomized clinical trial. *Pain*. 2021;162(9):2428-2435. doi:10.1097/j.pain.0000000000002234
- 2811 37. Nurko S, Saps M, Kossowsky J, others, Kelley JM. Effect of open-label placebo on children and adolescents with functional abdominal pain or irritable bowel syndrome. *JAMA Pediatrics*. 2022;176(4):349-356. doi:10.1001/jamapediatrics.2021.5750
- 2812 38. Benedetti F. Placebo effects: From the neurobiological paradigm to translational implications. *Neuron*. 2014;84(3):623-637.
- 2813 39. Frisaldi E, Shaibani A, Benedetti F. Understanding the mechanisms of placebo and nocebo effects. *Swiss Medical Weekly*. 2020;150:w20340. doi:10.4414/smw.2020.20340
- 2814 40. Carlino E, Frisaldi E, Benedetti F. Pain and the context. *Nature Reviews Rheumatology*. 2014;10(6):348-355. doi:10.1038/nrrheum.2014.17

- 2815 41. Carlino E, Piedimonte A, Benedetti F. Nature of the placebo and nocebo effect in relation to functional neurologic disorders. In: *Handbook of Clinical Neurology*. Vol 139.; 2016:597-606. doi:10.1016/B978-0-12-801772-2.00048-5
- 2816 42. Wager TD, Atlas LY, Lindquist MA, Roy M, Woo CW, Kross E. An fMRI-based neurologic signature of physical pain. *New England Journal of Medicine*. 2013;368(15):1388-1397. doi:10.1056/NEJMoa1204471
- 2817 43. Zunhammer M, Bingel U, Wager TD. Placebo effects on the neurologic pain signature: A meta-analysis of individual participant functional magnetic resonance imaging data. *JAMA Neurology*. 2018;75(11):1321-1330. doi:10.1001/jamaneurol.2018.2017
- 2818 44. Raskind MA, Peterson K, Williams T, et al. A trial of prazosin for combat trauma PTSD with nightmares in active-duty soldiers returned from Iraq and Afghanistan. *American Journal of Psychiatry*. 2013;170(9):1003-1010. doi:10.1176/appi.ajp.2013.12081133
- 2819 45. Holford NHG. Holford NHG and sheiner LB 'understanding the dose-effect relationship - clinical application of pharmacokinetic-pharmacodynamic models', Clin Pharmacokin 6:429-453 (1981) - the backstory. *The AAPS Journal*. 2011;13(4):662-664. doi:10.1208/s12248-011-9306-5
- 2820 46. Tilburt JC, Emanuel EJ, Kaptchuk TJ, Curlin FA, Miller FG. Prescribing 'placebo treatments': Results of national survey of US internists and rheumatologists. *BMJ*. 2008;337:a1938. doi:10.1136/bmj.a1938
- 2821 47. Finniss DG, Kaptchuk TJ, Miller F, Benedetti F. Biological, clinical, and ethical advances of placebo effects. *The Lancet*. 2010;375(9715):686-695. doi:10.1016/S0140-6736(09)61706-2
- 2822 48. Kirsch I. Response expectancy as a determinant of experience and behavior. *American Psychologist*. 1985;40(11):1189-1202. doi:10.1037/0003-066X.40.11.1189
- 2823 49. Senn S. Controversies concerning randomization and additivity in clinical trials. *Statistics in Medicine*. 2004;23(24):3729-3753. doi:10.1002/sim.2074
- 2824 50. Bingel U, Wanigasekera V, Wiech K, et al. The effect of treatment expectation on drug efficacy: Imaging the analgesic benefit of the opioid remifentanyl. *Science Translational Medicine*. 2011;3(70):70ra14. doi:10.1126/scitranslmed.3001244

- 2825 51. Colloca L, Lopiano L, Lanotte M, Benedetti F. Overt versus covert treatment for pain, anxiety, and Parkinson's disease. *The Lancet Neurology*. 2004;3(11):679-684.
- 2826 52. Rohsenow DJ, Marlatt GA. The balanced placebo design: Methodological considerations. *Addictive Behaviors*. 1981;6(2):107-122.
- 2827 53. International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use. ICH E9(R1) addendum on estimands and sensitivity analysis in clinical trials to the guideline on statistical principles for clinical trials. Published online 2019.
- 2828 54. Senn S. Statistical pitfalls of personalized medicine. *Nature*. 2018;563(7733):619-621.
- 2829 55. Daza EJ. Causal analysis of self-tracked time series data using a counterfactual framework for N-of-1 trials. *Methods of Information in Medicine*. 2018;57(S 01):e10-e21. doi:10.3414/ME16-02-0044
- 2830 56. Gärtner T, Schneider J, Arnrich B, Konigorski S. Comparison of Bayesian networks, G-estimation and linear models to estimate causal treatment effects in aggregated N-of-1 trials with carry-over effects. *BMC Medical Research Methodology*. 2023;23:191. doi:10.1186/s12874-023-02012-5
- 2831 57. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102.

2832

2833 *Rendered on 2026-06-15 at 09:21 PDT.*

2834 *Source: ~/prj/res/36-pmsimstats-ng/pmsimstats-ng/analysis/report/06-component-decomposition/r*

2026-06-15 09:21 PDT

76

~/prj/res/36-pmsimstats-ng/pmsimstats-ng/analysis/report/  
06-component-decomposition/report.Rmd